Ex 01 Estimating above-ground biomass

This report details three exercises undertaken to: 1) estimate aboveground biomass of individual trees using published allometric equations and a hectare expansion factor, 2) estimate combined aboveground biomass of trees at the plot level, 3) and to internally validate candidate models based on the limited data available. The current dataset was collected in Dumfries and Galloway region of southwest Scotland. A total of 167 trees were measured diameter at breast height (dbh > 1.3m), sampling populations in variable sized circular plots among four species: *Betula pendula, Betula pubescens, Sorbus aucuparia* and *Picea sitchensis.* Data analysis was conducted in R environment (v.2.1). The developed syntax is provided in the appendix.

Question 1-3; Identification of aboveground biomass equations

Biomass equations were selected based on four requirements: 1) location of study that is in closest proximity to current field site, 2) the relevant use of 'dbh' as the predictor in their biomass equation, 3) the study of relavant species, 4) and appropriate range of 'dbh' values assessed when fitting the biomass equation. Among eleven candiate equations considered, three models were based on logarithmic transformation of the variables. The following table presents the eleven candidate equations and their summary information:

Table 1. Aboveground biomass equations by tree species. Format provided under "Equation" and a, b, D, are parameter values. Index numbers indicating original publications listed below bibliography

Species (Index#)	Range of D	Equation	α	β	Country
(M1) *Betula pendula* (32)	/	$\alpha*D\beta$	0.2511	2.29	UK
(M2) *Betula pendula* (10)	2.9-30	$\alpha+\beta*\ln(D)$	-2.4166	2.4227	UK
(M3) *Betula pendula* (10)	2.9-26	$\alpha+\beta*\ln(D)$	-2.7584	2.6134	UK
(M4) *Betula pendula* (10)	3.3-16	$\alpha+\beta*\ln(D)$	-2.1625	2.3078	UK
(M5) *Betula pendula* (10)	3.5-23	$\alpha+\beta*\ln(D)$	-2.6423	2.4678	UK
(M6) *Betula pubescens* (35)	10-90	$\alpha*D\beta$	-2.162	2.3078	UK
(M7) *Betula pubescens* (35)	0.8-8.5	$\alpha*D\beta$	0.00029	2.50038	Sweden
(M8) *Picea sitchensis* (C16)	3-283	$\alpha+\beta*\ln(D)$	-3.0300	2.5567	US
(M9) *Picea sitchensis* (B04)	2-40	$\alpha*D\beta$	0.028	2.710	Ireland
(M10) *Sorbus aucuparia* (1)	3-64	$\alpha+\beta*\ln(D)$	-2.2118	2.4133	US
(M11) *Sorbus aucuparia* (2)	3-60	$\alpha+\beta*\ln(D)$	-2.9255	2.4109	US

Table 2 Descriptive statistics for all trees measured (n=167)

Variable	Count	Mean	SD	Median	SE	Range
Mean diameter (cm)						
Betula pendula	40	23.7	23.1	13.5	3.6	76.0
Betula pubescens	20	11.1	3.9	9.5	0.9	12.0
Picea sitchensis	66	19.7	6.5	20.0	0.8	26.0
Sorbus aucuparia	1	8.0	8.0	/	~~/~~	~~0.0~~

For silver brich, five biomass equations were available from the UK with similar range in values. To identify the best-fit equation, eleven models were fitted using response variables generated from the above equations. Descriptive analysis of these found highly dipsersed data, violation of homogeneity and non-normal distribution (Table 3). Plots were generated to examine residuals versus fitted values (Fig.1-8). Using non-linear least squared regression, models were compared by assessing the bias between predicted values and observed values. In table 4 below, the root mean square error of model residuals (RMSE) was presented in absolute terms. Models with highest RMSE values were omitted, providing final cut of eight candidate models.

Table 3. Descriptive statistics for response variables based on 7 biomass equations. #Shaprio-Wilks's p-values reporting signficant deviation from normal distribution at 0.05, 0.01, 0.001 leve;s as *, ** and **

Variable	Mean	SD	SE	Skewness	Kurtosis	p-value#
M1: SBI_agb_Z05.32	872.4	1533.7	242.5	1.73	1.53	***
M2: SBI_agb_Z05.52	531.8	962.0	152.1	1.77	1.68	***
M3: SBI_agb_Z05.53	824.5	1546.3	244.5	1.82	1.90	***
M4: SBI_agb_Z05.54	429.6	758.3	119.9	1.73	1.55	***
M6: PBI_agb_B68.01	35.1	29.1	6.5	1.19	0.22	***
M7: PBI_agb_Z05.40	0.2	0.1	0.0	1.25	0.39	***
M8: SS_agb_B04.01	111.3	83.4	10.3	0.71	-0.37	***
M9: SS_agb_C14.16	118.3	85.7	10.4	0.65	-0.48	**

For silver birch, M3 model was best fit to the current dataset. Bunce, who carried out inventory survey in the nearby Meathop Wood, derived this equation (Bunce 1968). Though, the range of dbh values appears less dispersed than this current dataset. For downy birch, some confusion arose. While model diagnostics indicated M7 was best fit, descriptive results suggested this equation was heavily conservative. Therefore, final equation was chosen from M6 model, which was also considerably significant. Interestingly, M6 was also derived from the same study carried out by Bunce (ibid.). As no biomass equation was provided for Rowan in the review conducted by Zianis et al (2005), a formula was adopted from a US-based study (Chojnacky, Heath, and Jenkins 2014). In this US-base study, two equations were available for Rowan. Between these, dbh ranges were considered and the more limited scale adopted (M11, dbh=3-60, Table 1). Between two equations for Sitka spruce, the models derived from a study in Ireland, which was best fit to the data, was chosen (Black et al. 2004).

Table 4. Parameter estimates (& standard error for biomass models: NLS-NLS5. SE: standard errors. RMSE: root mean square. Signficant estimates reported at the levels of 0.05, 0.01, 0.001 as *, **, ***.

Parameter	M1 (SBI)	M2 (SBI)	M3 (SBI)	M4 (SBI)
α	56.774 (9.901)***	52.076 (9.344)***	121.037 (22.548)***	29.817 (5.220)***
β	1.027 (0.011)***	0.684 (0.010)***	1.181 (0.025)***	0.5122 (0.006)***
RMSE	28.61	27.00	2.38	15.09

Parameter	M6 (PBI)	M7 (PBI)	M8 (SS)	M9 (SS)
α	7.185 (0.346)***	0.055 (0.003)***	38.760 (1.968)***	30.372 (1.467)***
β	-2.200 (0.059)***	0.002 (0.001)***	0.526 (0.006)***	0.495 (0.004)***
RMSE	0.14	0.00	1.98	1.47

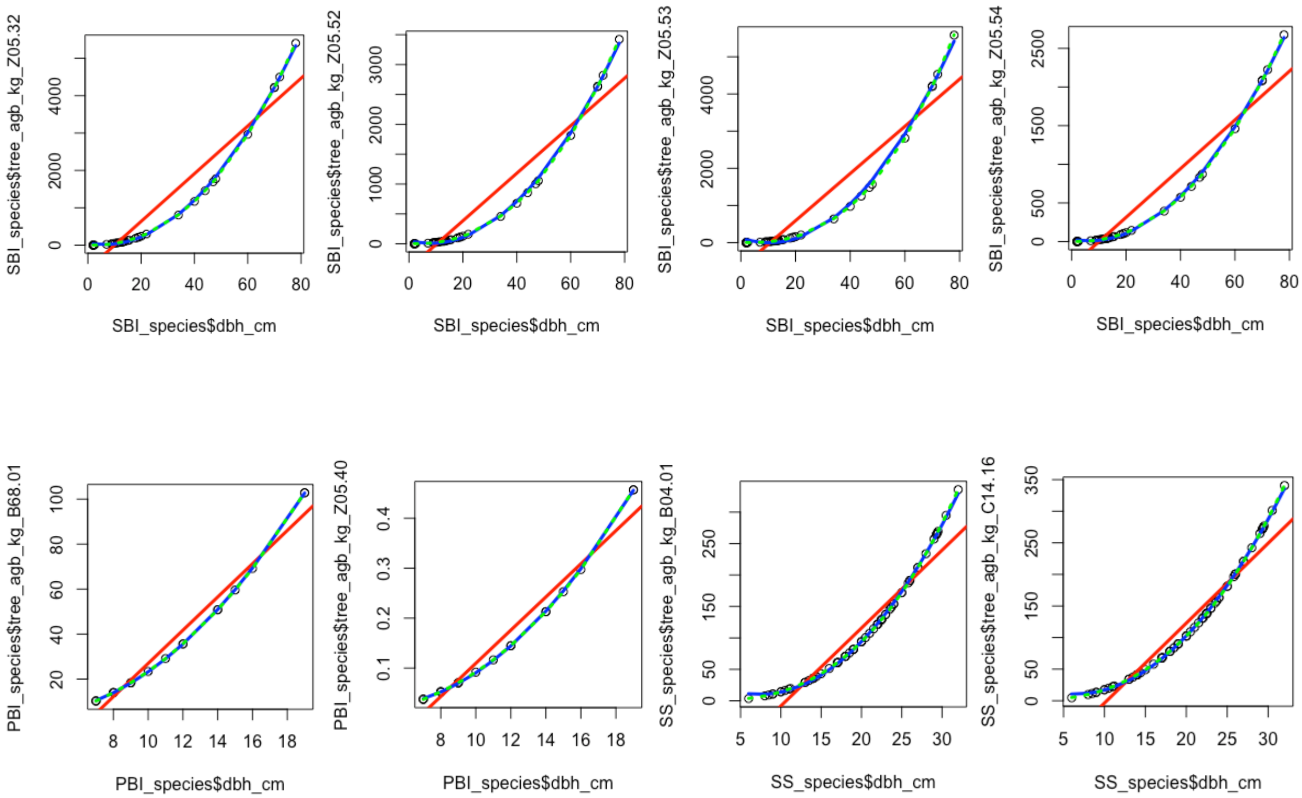
Hectare expansion factor was estimated using plot size as an attribute of the R-syntax. Using the relascope factor of 2, the radius of relascope plots was estimated using dbh, such that:

```
ForestInventoryRelascope_Rversion$plot_radius_m <- (35*ForestInventoryRelascope_Rversion$dbh_cm/100)
ForestInventoryRelascope_Rversion$hef <- 10000 / (pi*ForestInventoryRelascope_Rversion$plot_radius_m^2)
```

Using a hectare area factor, the absolute contribution of basal area (BA_m^2_ha^{-1}) and aboveground biomass (AGB_Mg_ha^{-1}) per hectare of individual trees and plots (Table 5). Having completed the exercise, it is recommended that the workflow of this analysis would benefit from beginning with hectare area factor before generating other results (see syntax in appendix)

plot	BA_m2_ha	AGB_Mg_ha
1	22.4062315	86.93626169
2	33.2564659	112.5917576
3	34.13074419	260.2069904

Figures 1-8 Scatterplot and regression lines of the relation between species dbh and the calculated aboveground biomass; Linear (red curve) vs nonlinear algorithmic regression lines (blue curve).



Bibliography

Black, Kevin et al. 2004. "Improved Estimates of Biomass Expansion Factors for Sitka Spruce." *Irish Forestry*.

Bunce, R G H. 1968. "Biomass and Production of Trees in a Mixed Deciduous Woodland: I. Girth and Height as Parameters for the Estimation of Tree Dry Weight." *The Journal of Ecology*: 759–75.

Chojnacky, David C, Linda S Heath, and Jennifer C Jenkins. 2014. "Updated Generalized Biomass Equations for North American Tree Species." *Forestry* 87(1): 129–51.

Zianis, Dimitris, Petteri Muukkonen, Raisa Mäkipää, and Maurizio Mencuccini. 2005. *Biomass and Stem Volume Equations for Tree Species in Europe*. FI.

Appendix I Biomass equations

- M1 Hughes, M. K. (1971). Tree biocontent, net production and litter fall in a deciduous woodland. *Oikos*, 62-73
- M2 Bunce, R. G. H. (1968). Biomass and production of trees in a mixed deciduous woodland: I. Girth and height as parameters for the estimation of tree dry weight. *The Journal of Ecology*, 759-775.
- M3 Bunce, R. G. H. (1968). Biomass and production of trees in a mixed deciduous woodland: I. Girth and height as parameters for the estimation of tree dry weight. *The Journal of Ecology*, 759-775.
- M4 Bunce, R. G. H. (1968). Biomass and production of trees in a mixed deciduous woodland: I. Girth and height as parameters for the estimation of tree dry weight. *The Journal of Ecology*, 759-775.
- M5 Bunce, R. G. H. (1968). Biomass and production of trees in a mixed deciduous woodland: I. Girth and height as parameters for the estimation of tree dry weight. *The Journal of Ecology*, 759-775.
- M6 Johansson, T. (1999). Biomass equations for determining fractions of pendula and pubescent birches growing on abandoned farmland and some practical implications. *Biomass and bioenergy*, *16*(3), 223-238.
- M7 Johansson, T. (1999). Biomass equations for determining fractions of pendula and pubescent birches growing on abandoned farmland and some practical implications. *Biomass and bioenergy*, *16*(3), 223-238.
- M8 Chojnacky, D. C., Heath, L. S., & Jenkins, J. C. (2014). Updated generalized biomass equations for North American tree species. *Forestry*, *87*(1), 129-151.
- M9 Black, K., Tobin, B., Saiz, G., Byrne, K. A., & Osborne, B. (2004). Improved estimates of biomass expansion factors for Sitka spruce. *Irish Forestry*.
- M10 Chojnacky, D. C., Heath, L. S., & Jenkins, J. C. (2014). Updated generalized biomass equations for North American tree species. *Forestry*, *87*(1), 129-151.
- M11 Chojnacky, D. C., Heath, L. S., & Jenkins, J. C. (2014). Updated generalized biomass equations for North American tree species. *Forestry*, *87*(1), 129-151.

Appendix II – R_syntax

```
###### Inventory Monitoring and Assessment ######
######  Assignment 1 - Tree Allometry ###########
########### S.Murphy -08-03-2020 ################

## Load data ##
## Compute HEF per plot data separately & calculate relascope plots using
 a:B ratio of 1:35 ##
ForestInventoryFixedSize_Rversion$hef <- 10000 /
 (pi*ForestInventoryFixedSize_Rversion$plot_radius_m^2)
ForestInventoryConcentric_Rversion$hef <- 10000 /
 (pi*ForestInventoryConcentric_Rversion$plot_radius_m^2)
ForestInventoryKtree_Rversion$hef <- 10000 /
 (pi*ForestInventoryKtree_Rversion$plot_radius_m^2)
ForestInventoryRelascope_Rversion$plot_radius_m <-
 (35*ForestInventoryRelascope_Rversion$dbh_cm/100)
ForestInventoryRelascope_Rversion$hef <- 10000 /
 (pi*ForestInventoryRelascope_Rversion$plot_radius_m^2)

## Merge datasets ##
total_trees <- rbind(ForestInventoryFixedSize_Rversion,
 ForestInventoryConcentric_Rversion, ForestInventoryKtree_Rversion,
```

```
 ForestInventoryRelascope_Rversion)

## Generate basal area per tree using dbh ##
total_trees$tree_ba_m <- ((total_trees$dbh_cm)/200)^2*3.147

## split data subsets ##
total_trees$species <- as.factor(total_trees$species)
total_trees$species_code <- recode(total_trees$species, 'SS'=1, 'PBI'=2,
 'SBI'=3, 'ROW'=4)
SS_species <- total_trees[ which(total_trees$species_code==1),]
PBI_species <- total_trees[ which(total_trees$species_code==2),]
SBI_species <- total_trees[ which(total_trees$species_code==3),]
ROW_species <- total_trees[ which(total_trees$species_code==4),]

## Generate "abg" response variables using selected biomass equations ##
## Equations used: zianis05.32, zianis05.52, zianis05.53, zianis05.54
## zianis05.55, zianis05.54, bunce68.01, zianis05.40, chojnacky.C14.01.172
## chojnacky.C14.02.172, black.B04.01, chojnacky.C14.16 - see bilbiography
 ##
SBI_species$tree_agb_kg_Z05.32 <- (((SBI_species$dbh_cm)^2.29)*0.2511)
SBI_species$tree_agb_kg_Z05.52 <- (exp(-2.4166
 +2.4227*log(SBI_species$dbh_cm)))
SBI_species$tree_agb_kg_Z05.53 <- (exp(-2.7584
 +2.6134*log(SBI_species$dbh_cm)))
SBI_species$tree_agb_kg_Z05.54 <- (exp(-2.1625
 +2.3078*log(SBI_species$dbh_cm)))
SBI_species$tree_agb_kg_Z05.55 <- (exp(-2.6423
 +2.4678*log(SBI_species$dbh_cm)))
PBI_species$tree_agb_kg_B68.01 <- (exp(-2.162 +
 2.3078*log(PBI_species$dbh_cm)))
PBI_species$tree_agb_kg_Z05.40 <- (((PBI_species$dbh_cm)^2.50038)*0.00029)
ROW_species$tree_agb_kg_C14.01 <- (exp(-2.9255 +
 2.4109*log(ROW_species$dbh_cm)))
ROW_species$tree_agb_kg_C14.02 <- (exp(-2.2118 +
 2.4133*log(ROW_species$dbh_cm)))
SS_species$tree_agb_kg_B04.01 <- (((SS_species$dbh_cm)^2.71)*0.028)
SS_species$tree_agb_kg_C14.16 <- (exp(-3.030 +
 2.5567*log(SS_species$dbh_cm)))

## Analysis; Run descriptives and check distrubtion of explanatory/response
 variables ##
describe(SBI_species$dbh_cm)
describe(PBI_species$dbh_cm)
describe(SS_species$dbh_cm)
```

```
describe(ROW_species$dbh_cm)

describe(SBI_species$tree_ba_m)
describe(PBI_species$tree_ba_m)
describe(SS_species$tree_ba_m)
describe(ROW_species$tree_ba_m)

describe(SBI_species$tree_agb_kg_Z05.32)
describe(SBI_species$tree_agb_kg_Z05.52)
describe(SBI_species$tree_agb_kg_Z05.53)
describe(SBI_species$tree_agb_kg_Z05.54)
describe(SBI_species$tree_agb_kg_Z05.55)
describe(PBI_species$tree_agb_kg_B68.01)
describe(PBI_species$tree_agb_kg_Z05.40)
describe(ROW_species$tree_agb_kg_C14.01)
describe(ROW_species$tree_agb_kg_C14.02)
describe(SS_species$tree_agb_kg_B04.01)
describe(SS_species$tree_agb_kg_C14.16)

plot(SBI_species$dbh_cm, SBI_species$tree_agb_kg_Z05.32)
abline(predict1.z05.32, lwd=3, col="red")
predict2.z05.32 <- lm(SBI_species$tree_agb_kg_Z05.32 ~ SBI_species$dbh_cm +
 I(SBI_species$dbh_cm^2))
lines(smooth.spline(SBI_species$dbh_cm, predict(predict2.z05.32)), lwd=3,
 col="blue")
predict3.z05.32 <- lm(SBI_species$tree_agb_kg_Z05.32 ~ SBI_species$dbh_cm +
 I(SBI_species$dbh_cm^2) + I(SBI_species$dbh_cm^3))
lines(smooth.spline(SBI_species$dbh_cm, predict(predict3.z05.32)),
 col="green", lwd=3, lty=3)


## Analysis; Check models for distribution, homoskedasticity##
## Compare linear/non-linear models ##
## Eq z05.32 ##

predict1.z05.32 <- lm(SBI_species$tree_agb_kg_Z05.32 ~ SBI_species$dbh_cm)
plot(SBI_species$tree_agb_kg_Z05.32, resid(predict1.z05.32))
plot(fitted(predict1.z05.32), resid(predict1.z05.32))
qqnorm(rstandard(predict1.z05.32))

plot(SBI_species$dbh_cm, SBI_species$tree_agb_kg_Z05.32)
abline(predict1.z05.32, lwd=3, col="red")
predict2.z05.32 <- lm(SBI_species$tree_agb_kg_Z05.32 ~ SBI_species$dbh_cm +
 I(SBI_species$dbh_cm^2))
```

```
lines(smooth.spline(SBI_species$dbh_cm, predict(predict2.z05.32)), lwd=3,
 col="blue")
predict3.z05.32 <- lm(SBI_species$tree_agb_kg_Z05.32 ~ SBI_species$dbh_cm +
 I(SBI_species$dbh_cm^2) + I(SBI_species$dbh_cm^3))
lines(smooth.spline(SBI_species$dbh_cm, predict(predict3.z05.32)),
 col="green", lwd=3, lty=3)

## Eq z05.52 ##
predict1.z05.52 <- lm(SBI_species$tree_agb_kg_Z05.52 ~ SBI_species$dbh_cm)
plot(SBI_species$tree_agb_kg_Z05.52, resid(predict1.z05.52))
plot(fitted(predict1.z05.52), resid(predict1.z05.52))
qqnorm(rstandard(predict1.z05.52))

plot(SBI_species$dbh_cm, SBI_species$tree_agb_kg_Z05.52)
abline(predict1.z05.52, lwd=3, col="red")
predict2.z05.52 <- lm(SBI_species$tree_agb_kg_Z05.52 ~ SBI_species$dbh_cm +
 I(SBI_species$dbh_cm^2))
lines(smooth.spline(SBI_species$dbh_cm, predict(predict2.z05.52)), lwd=3,
 col="blue")
predict3.z05.52 <- lm(SBI_species$tree_agb_kg_Z05.52 ~ SBI_species$dbh_cm +
 I(SBI_species$dbh_cm^2) + I(SBI_species$dbh_cm^3))
lines(smooth.spline(SBI_species$dbh_cm, predict(predict3.z05.52)),
 col="green", lwd=3, lty=3)

## Eq z05.53 ##
predict1.z05.53 <- lm(SBI_species$tree_agb_kg_Z05.53 ~ SBI_species$dbh_cm)
plot(SBI_species$tree_agb_kg_Z05.53, resid(predict1.z05.53))
plot(fitted(predict1.z05.53), resid(predict1.z05.53))
qqnorm(rstandard(predict1.z05.53))

plot(SBI_species$dbh_cm, SBI_species$tree_agb_kg_Z05.53)
abline(predict1.z05.53, col="red", lwd=3)
predict2.z05.53 <- lm(SBI_species$tree_agb_kg_Z05.53 ~ SBI_species$dbh_cm +
 I(SBI_species$dbh_cm^2))
lines(smooth.spline(SBI_species$dbh_cm, predict(predict2.z05.53)), lwd=3,
 col="blue")
predict3.z05.53 <- lm(SBI_species$tree_agb_kg_Z05.53 ~ SBI_species$dbh_cm +
 I(SBI_species$dbh_cm^2) + I(SBI_species$dbh_cm^3))
lines(smooth.spline(SBI_species$dbh_cm, predict(predict3.z05.53)),
 col="green", lwd=3, lty=3)

## Eq z05.54 ##
predict1.z05.54 <- lm(SBI_species$tree_agb_kg_Z05.54 ~ SBI_species$dbh_cm)
plot(SBI_species$tree_agb_kg_Z05.54, resid(predict1.z05.54))
```

```
plot(fitted(predict1.z05.54), resid(predict1.z05.54))
qqnorm(rstandard(predict1.z05.54))

plot(SBI_species$dbh_cm, SBI_species$tree_agb_kg_Z05.54)
abline(predict1.z05.54, col="red", lwd=3)
predict2.z05.54 <- lm(SBI_species$tree_agb_kg_Z05.54 ~ SBI_species$dbh_cm +
 I(SBI_species$dbh_cm^2))
lines(smooth.spline(SBI_species$dbh_cm, predict(predict2.z05.54)), lwd=3,
 col="blue")
predict3.z05.54 <- lm(SBI_species$tree_agb_kg_Z05.54 ~ SBI_species$dbh_cm +
 I(SBI_species$dbh_cm^2) + I(SBI_species$dbh_cm^3))
lines(smooth.spline(SBI_species$dbh_cm, predict(predict3.z05.54)),
 col="green", lwd=3, lty=3)

## Eq z05.55 ##
predict1.z05.55 <- lm(SBI_species$tree_agb_kg_Z05.55 ~ SBI_species$dbh_cm)
plot(SBI_species$tree_agb_kg_Z05.55, resid(predict1.z05.55))
plot(fitted(predict1.z05.55), resid(predict1.z05.55))
qqnorm(rstandard(predict1.z05.55))

plot(SBI_species$dbh_cm, SBI_species$tree_agb_kg_Z05.55)
abline(predict1.z05.55, col="red", lwd=3)
predict2.z05.55 <- lm(SBI_species$tree_agb_kg_Z05.55 ~ SBI_species$dbh_cm +
 I(SBI_species$dbh_cm^2))
lines(smooth.spline(SBI_species$dbh_cm, predict(predict2.z05.55)), lwd=3,
 col="blue")
predict3.z05.55 <- lm(SBI_species$tree_agb_kg_Z05.55 ~ SBI_species$dbh_cm +
 I(SBI_species$dbh_cm^2) + I(SBI_species$dbh_cm^3))
lines(smooth.spline(SBI_species$dbh_cm, predict(predict3.z05.55)),
 col="green", lwd=3, lty=3)

## Eq B68.01 ##
predict1.B68.01 <- lm(PBI_species$tree_agb_kg_B68.01 ~ PBI_species$dbh_cm)
plot(PBI_species$tree_agb_kg_B68.01, resid(predict1.B68.01))
plot(fitted(predict1.B68.01), resid(predict1.B68.01))
qqnorm(rstandard(predict.B68.01))

plot(PBI_species$dbh_cm, PBI_species$tree_agb_kg_B68.01)
abline(predict1.B68.01, col="red", lwd=3)
predict2.B68.01 <- lm(PBI_species$tree_agb_kg_B68.01 ~ PBI_species$dbh_cm +
 I(PBI_species$dbh_cm^2))
lines(smooth.spline(PBI_species$dbh_cm, predict(predict2.B68.01)), lwd=3,
 col="blue")
predict3.B68.01 <- lm(PBI_species$tree_agb_kg_B68.01 ~ PBI_species$dbh_cm +
```

```
 I(PBI_species$dbh_cm^2) + I(PBI_species$dbh_cm^3))
lines(smooth.spline(PBI_species$dbh_cm, predict(predict3.B68.01)),
 col="green", lwd=3, lty=3)

## Eq Z05.40 ##
predict1.z05.40 <- lm(PBI_species$tree_agb_kg_Z05.40 ~ PBI_species$dbh_cm)
plot(PBI_species$tree_agb_kg_Z05.40, resid(predict1.z05.40))
plot(fitted(predict1.z05.40), resid(predict1.z05.40))
qqnorm(rstandard(predict1.z05.40))

plot(PBI_species$dbh_cm, PBI_species$tree_agb_kg_Z05.40)
abline(predict1.z05.40, col="red", lwd=3)
predict2.z05.40 <- lm(PBI_species$tree_agb_kg_Z05.40 ~ PBI_species$dbh_cm +
 I(PBI_species$dbh_cm^2))
lines(smooth.spline(PBI_species$dbh_cm, predict(predict2.z05.40)), lwd=3,
 col="blue")
predict3.z05.40 <- lm(PBI_species$tree_agb_kg_Z05.40 ~ PBI_species$dbh_cm +
 I(PBI_species$dbh_cm^2) + I(PBI_species$dbh_cm^3))
lines(smooth.spline(PBI_species$dbh_cm, predict(predict3.z05.40)),
 col="green", lwd=3, lty=3)

## Eq B04.01 ##
predict1.B04.01 <- lm(SS_species$tree_agb_kg_B04.01 ~ SS_species$dbh_cm)
plot(SS_species$tree_agb_kg_B04.01, resid(predict1.B04.01))
plot(fitted(predict1.B04.01), resid(predict1.B04.01))
qqnorm(rstandard(predict1.B04.01))

plot(SS_species$dbh_cm, SS_species$tree_agb_kg_B04.01)
abline(predict1.B04.01, col="red", lwd=3)
predict2.B04.01 <- lm(SS_species$tree_agb_kg_B04.01 ~ SS_species$dbh_cm +
 I(SS_species$dbh_cm^2))
lines(smooth.spline(SS_species$dbh_cm, predict(predict2.B04.01)), lwd=3,
 col="blue")
predict3.B04.01 <- lm(SS_species$tree_agb_kg_B04.01 ~ SS_species$dbh_cm +
 I(SS_species$dbh_cm^2) + I(SS_species$dbh_cm^3))
lines(smooth.spline(SS_species$dbh_cm, predict(predict3.B04.01)),
 col="green", lwd=3, lty=3)

## Eq C14.16 ##
predict1_C14.16 <- lm(SS_species$tree_agb_kg_C14.16 ~ SS_species$dbh_cm)
plot(SS_species$tree_agb_kg_C14.16, resid(predict1_C14.16))
plot(fitted(predict1_C14.16), resid(predict1_C14.16))
qqnorm(rstandard(predict1_C14.16))
```

```
plot(SS_species$dbh_cm, SS_species$tree_agb_kg_C14.16)
abline(predict1_C14.16, col="red", lwd=3)
predict2_C14.16 <- lm(SS_species$tree_agb_kg_C14.16 ~ SS_species$dbh_cm +
 I(SS_species$dbh_cm^2))
lines(smooth.spline(SS_species$dbh_cm, predict(predict2_C14.16)), lwd=3,
 col="blue")
predict3_C14.16 <- lm(SS_species$tree_agb_kg_C14.16 ~ SS_species$dbh_cm +
 I(SS_species$dbh_cm^2) + I(SS_species$dbh_cm^3))
lines(smooth.spline(SS_species$dbh_cm, predict(predict3_C14.16)),
 col="green", lwd=3, lty=3)

## Compare model resuls ##
summary(predict2.z05.32)
summary(predict2.z05.52)
summary(predict2.z05.53)
summary(predict2.z05.54)
summary(predict2.z05.55)
summary(predict2.B68.01)
summary(predict2.z05.40)
summary(predict2.B04.01)
summary(predict2_C14.16)

## split fixed_size_data to estiamte plot level BA & AGB##
sapply(ForestInventoryFixedSize_Rversion, class)
ForestInventoryFixedSize_Rversion$species <-
 as.factor(ForestInventoryFixedSize_Rversion$species)
ForestInventoryFixedSize_Rversion$species_code <-
 recode(ForestInventoryFixedSize_Rversion$species, 'SS'=1, 'PBI'=2, 'SBI'=3,
 'ROW'=4)
SS_species <- ForestInventoryFixedSize_Rversion[
 which(ForestInventoryFixedSize_Rversion$species_code==1),]
PBI_species <- ForestInventoryFixedSize_Rversion[
 which(ForestInventoryFixedSize_Rversion$species_code==2),]
SBI_species <- ForestInventoryFixedSize_Rversion[
 which(ForestInventoryFixedSize_Rversion$species_code==3),]
ROW_species <- ForestInventoryFixedSize_Rversion[
 which(ForestInventoryFixedSize_Rversion$species_code==4),]

## generate tree_agb_kg using selected biomass equations ##
SBI_species$tree_agb_kg <- (exp(-2.7584 +2.6134*log(SBI_species$dbh_cm)))
PBI_species$tree_agb_kg <- (((PBI_species$dbh_cm)^2.50038)*0.00029)
ROW_species$tree_agb_kg <- (exp(-2.9255 + 2.4109*log(ROW_species$dbh_cm)))
SS_species$tree_agb_kg <- (exp(-3.030 + 2.5567*log(SS_species$dbh_cm)))
```

```
# merge fixed_size_datasets
fixedplot_merged <- rbind(PBI_species, SBI_species, ROW_species, SS_species)

fixedplot_merged$tree_ba_m <- ((fixedplot_merged$dbh_cm)/200)^2*3.147
fixedplot_merged$BA_m <- (fixedplot_merged$tree_ba_m*fixedplot_merged$hef)
fixedplot_merged$AGB_Mg_ha <-
 (fixedplot_merged$tree_agb_kg*fixedplot_merged$hef)

fixedplot_merged %>%
  group_by(plot) %>%
  summarise(sumplot_AGB = sum(fixedplot_merged$BA_m, na.rm = TRUE))
```

Introduction:

Page one of this report addresses the seven questions provided in this exercise protocol (Q1-7). These were addressed using analysis of vegetation classification based on a false colour infrared image of the Clocaenog Forest of Denbighshire, North Wales. Page two of this report discusses the four steps of analysis taken, which included 1) exploration and segmentation of image data, 2) non-supervised classification of extracted image data using ISODATA algorithm 3) supervised classification of image data using maximum likelihood classifier (MLC), support vector machine methods (SVM) and random tree classifier (RTC), and 4) a post-classification estimation of accuracy between the three supervised classification. This step used the unsupervised ISO clustering as ground truth data. However due to time constraints and arc issues, it was not possible to complete reclassification of iso clustering necessary to match the 9 training sample, thereby producing thereby producing negative kappa coefficients and poor classification accuracies. These errors are reported in final section.

Q1-3 Explain why there are different ranges of data between raster files of predictor variables, discuss any significant correlation between bands, and what meaning different bands hold regarding land cover.

Regarding data range of each raster file, there are two possible answers. On the one hand, these bands comprise varying ranges of frequencies along the electromagnetic spectrum. On the other, data range may refer to minimum and maximum values of these 8-bit grayscale raster files (0=black, 255=white). Regarding the former, the near infrared band (Band 1) senses images in the longest spectral range (0.7 - 100μm), the red band (Band 2) senses images in the range (0.500 - 0.578μm) and the green band (Band 3) senses in the shortest range (0.620 - 0.7μm). Regarding the latter, data ranges generated by the band statistics tool found that bands 1 and 3 ranges from 0 and 255, and from 3 to 255 for band 2. Greatest variance was observed in NIR band (SD = 38.1116). Highly significant correlation was observed between band 2 and band 3 ($R = 0.974$). Due to the different characteristics of electromagnetic energies and the different interactions they produces with surface targets and atmospheric interference, these bands provide different information regarding land cover. Among visible wavelengths, chlorophyll most strongly absorbs radiation in the red wavelengths. In contrast, healthy leaves are highly reflecting of NIR wavelengths. As a result, NIR bands are used as key indicators of vegetation change (Treitz and Howarth 1999) and vegetation indices (Elvidge and Chen 1995; Milchunas 1998). If considering grayscale data ranges, a possible reason for red band having shorter range could relate to the fine resolution of this image (0.5m), which means that wavelengths are narrower. This effect can limit spatial resolution and texture relations between pixel values, which may limit classification algorithms (Chakraborty et al. 2013) or obscure lower values (black pixels) from red band wavelengths.

Q4-7 What types of land use can we distinguish, is it possible to plan feeling/planting activities with these methods, or distinguish tree species and stand compositions, and which species are separable?

With this kind of aerial imagery, it is possible to distinguish between broadleaves and conifers, and between stand compositions and age classes (Jensen, Qiu, and Ji 1999; Pilger, Peddle, and Luther 2002; Roy, Sharma, and Jain 1996; Spanner et al. 1990). However, identifying between evergreen conifer species remains difficult as their spectral response is very similar (Coleman, Gudapati, and Derrington 1990; Niemann 1995). Classification of Clocaenog identified a total of 10 land cover types relating to farming and forestry: bare land, crops, pasture, thick scrub, sapling plantations, broadleaf woodlands, young evergreen woodlands, mature evergreen woodlands and larch woodlands (Table 3). Image classification can also be used to help plan felling and afforestation activities (Gemmell 1995), or to map forests according to plan stress (Mutanga and Skidmore 2004; Xu et al. 2019). Studies of conifer forests, using bands 3, 4, and 5 of Landsat 7, found stand development returned highest accuracy of classification among all stand type parameters (Günlü et al. 2008). Others have used R-NIR wavelengths (red edge = 0.68 – 0.75μm) to capture mesophyll cells and changes in needle structure to map deciduous conifer species (Touzel 2013). This was similar in the current classification of larch stands in Clocaenog, which observed highest mean grayscale value in band 1 (156.84, NIR) In addition, accuracy of classifcation can be improved by increasing spectral detail and integrating textural information through noise-reduction parameters (Pilger, Peddle, and Luther 2002). To avoid over-segmentation in such processes, sequential classification and object-based image analysis is recommended (Blaschke 2010; Geneletti and Gorte 2003; Zou et al 2016).

Step 1 Exploration and segmentation of multispectral data

To check for classification error through unequal spread of means, the distributions of predictor variables were assessed (A; Table 1). Using 'SegmentMeanShift' tool in arc, the image data was partitioned into coarsely homogenous clusters according to spectral similarities. This tool is based on the kernel density function (Jain, Murty, and Flynn 1999).

Table 1 Descriptive statistics of band 1 (NIR), band 2 (Red),, band 3 (Green)

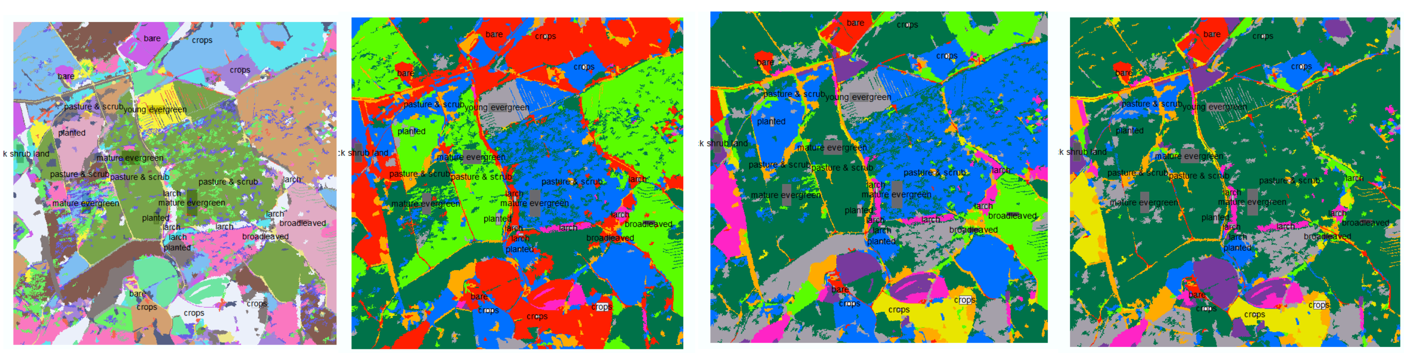
	Mean	SD	Min	Max	Correlation matrix		
					Band 1	Band 2	Band 3
Band 1	115.63	38.11	0.00	255.00	1.0000	0.7225	0.7879
Band 2	106.28	35.29	3.00	255.00	0.7225	1.0000	0.9738
Band 3	90.37	29.00	0.00	255.00	0.7879	0.9738	1.0000

Step 2 Non-supervised classification of image data

Using segmented data from above, pixels were clustered using the isodata algorithm (interactive self-organising data analysis technique, Chakraborty et al. 2015; 2013; 2012; Chakraborty, Singh, and Dutta 2017; Thakur and Chakraborty 2019). Basd on the k-means algorithm or the minimum Euclidean distance, isoclustering process iteratively assigns pixels to mearest classes that are recalculated according to their members locations (Gordon and Henderson 1977; Mao and Jain 1996). Class mean vectors are recalculated and pixels reassigned until there is limited change between iterations (Jain, Murty, and Flynn 1999) and end-point class signatures are defined and pixels are classified according to the minimum distance classifer. From these results, additional sub-classes were obserbevd that were not noted by the 7 training sites from the exercise folder nor in the forestry commission shape file (Fig 4). In this image, three new training sites were added: 'thick shrub lands' (N303024.7, E354394.8'), 'young evergreen' (N303429.0, E354244.7), and 'mature evergreen' (N303654.9, E354238.4).

Step 3 Supervised classification image data

Three classification methods were applied. Each of the MLC, RTC and SVM classifiers were trained to the 9 training samples, before the three algorithms were run across through the image data. The MLC tool calculated probabilities of pixels belonging to each training sample before reassigning them to new classes (Figure 5). Similar to pixel-based classification, the RTC and SVM methods used object-oriented feature extractions to classify segments or super pixel within the data. Visually this represented a coarser classification that MLC (Figure 5-8). To review the distribution of these classes per tool, spectral signatures are provided in appendix.



Figs. 4-8. Unsupervised ISO and supervised MLC, RTC and SVM classifications of image data, respectively

Step 4 Accuracy assessment of classifications

Although accuracy estimations were flawed due to mismatching of class numbers between ground truth data (ISO image) and supervised classification systems, it was curious that SVM (-1.0%) and RTC (-1.1%) proved more accurate than MLS (-2.1%). Research highlights machine learning algorithms for land-cover classifications (Xie et al. 2019), though choice of tool depends on site and classification system put in place (Lu and Weng 2007).

Table 2. Spectral signatures and accuracy estimates of land-cover classes of MLC, SVM, RTC

Classifier	# Classes	User accuracy	Kappa agreement	Significant class
MLC	9	0.0176	-0.0207	C6 (0.0909)
RTC	9	0.0275	-0.0112	C8 (0.2727)
SVM	9	0.0246	-0.0060	C3 (0.1000)

Bibliography

Blaschke, T. 2010. “No TitlObject Based Image Analysis for Remote Sensing. E.” *ISPRS Journal of Photogrammetry and Remote Sensing, 65(2), pp. 1-16.Blaschke, T., 2010. Object Based Image Analysis for Remote Sensing. ISPRS Journal of Photogrammetry and Remote Sensing,, pp. 1-16.*

Chakraborty, D, V M Chowdhary, D Dutta, and J R Sharma. 2013. “Classification of High Spatial Resolution Image Using Multi Circular Local Binary Pattern and Variance.” *International Journal of Electronics and Computer Engineering* 4(6): 1648–53.

Coleman, T L, L Gudapati, and J Derrington. 1990. “Monitoring Forest Plantations Using Landsat Thematic Mapper Data.” *Remote Sensing of Environment* 33(3): 211–21.

Elvidge, Christopher D, and Zhikang Chen. 1995. “Comparison of Broad-Band and Narrow-Band Red and near-Infrared Vegetation Indices.” *Remote sensing of environment* 54(1): 38–48.

Gemmell, F M. 1995. “Effects of Forest Cover, Terrain, and Scale on Timber Volume Estimation with Thematic Mapper Data in a Rocky Mountain Site.” *Remote Sensing of Environment* 51(2): 291–305.

Geneletti, D, and B G H Gorte. 2003. “A Method for Object-Oriented Land Cover Classification Combining Landsat TM Data and Aerial Photographs.” *International Journal of Remote Sensing* 24(6): 1273–86.

Gordon, A D, and J T Henderson. 1977. “An Algorithm for Euclidean Sum of Squares Classification.” *Biometrics*: 355–62.

Günlü, Alkan et al. 2008. “Estimation of Stand Type Parameters and Land Cover Using Landsat-7 ETM Image: A Case Study from Turkey.” : 2509–25.

Jain, Anil K, M Narasimha Murty, and Patrick J Flynn. 1999. “Data Clustering: A Review.” *ACM computing surveys (CSUR)* 31(3): 264–323.

Jensen, J R, Fang Qiu, and Minhe Ji. 1999. “Predictive Modelling of Coniferous Forest Age Using Statistical and Artificial Neural Network Approaches Applied to Remote Sensor Data.” *International Journal of Remote Sensing* 20(14): 2805–22.

Lu, Dengsheng, and Qihao Weng. 2007. “A Survey of Image Classification Methods and Techniques for Improving Classification Performance.” *International journal of Remote sensing* 28(5): 823–70.

Mao, Jianchang, and Anil K Jain. 1996. “A Self-Organizing Network for Hyperellipsoidal Clustering (HEC).” *Ieee transactions on neural networks* 7(1): 16–29.

Milchunas, S W. 1998. “Biomass Estimation on Grazed and Ungrazed Rangelands Using Spectral Indices.” *International journal of remote sensing* 19(3): 427–38.

Mutanga, Onisimo, and Andrew K Skidmore. 2004. “Narrow Band Vegetation Indices Overcome the Saturation Problem in Biomass Estimation.” *International Journal of Remote Sensing* 25(19): 3999–4014.

Niemann, K Olaf. 1995. “Remote Sensing of Forest Stand Age Using Airborne Spectrometer Data.” *Photogrammetric Engineering and Remote Sensing* 61(9): 1119–28.

Pilger, N, Derek R Peddle, and Joan E Luther. 2002. “Estimation of Forest Cover Type and Structure from

Landsat TM Imagery Using a Canopy Reflectance Model for Biomass Mapping in Western Newfoundland." In *IEEE International Geoscience and Remote Sensing Symposium*, , 1324–26.

Roy, P S, K P Sharma, and Abhineet Jain. 1996. "Stratification of Density in Dry Deciduous Forest Using Satellite Remote Sensing Digital Data—An Approach Based on Spectral Indices." *Journal of Biosciences* 21(5): 723–34.

Spanner, Michael A, Lars L Pierce, David L Peterson, and Steven W Running. 1990. "Remote Sensing of Temperate Coniferous Forest Leaf Area Index The Influence of Canopy Closure, Understory Vegetation and Background Reflectance." *TitleREMOTE SENSING* 11(1): 95–111.

Touzel, Steven Joseph. 2013. "Mapping the Distribution and Abundance of Western Larch (Larix Occidentalis Nutt.) with Multi-Temporal Satellite Imagery and Gradient Modeling."

Treitz, Paul M, and Philip J Howarth. 1999. "Hyperspectral Remote Sensing for Estimating Biophysical Parameters of Forest Ecosystems." *Progress in Physical Geography* 23(3): 359–90.

Xiaoliang ZOU, Guihua ZHAO, Jonathan LI, Yuanxi YANG, ong FANG. 2016. "OBJECT BASED IMAGE ANALYSIS COMBINING HIGH SPATIAL RESOLUTION IMAGERY AND LASER POINT CLOUDS FOR URBAN LAND COVER." *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences, Volume XLI-B3, 2016*.

Xie, Zhuli et al. 2019. "Classification of Land Cover, Forest, and Tree Species Classes with ZiYuan-3 Multispectral and Stereo Data." *Remote Sensing* 11(2): 164.

Xu, Nianxu et al. 2019. "Analysis of Vegetation Red Edge with Different Illuminated/Shaded Canopy Proportions and to Construct Normalized Difference Canopy Shadow Index." *Remote Sensing* 11(10): 1192.

Yeşil, A.; Musaoğlu, N.; Kaya, Ş.; Çoşkun, G.; Asan, Ü.; Örmeci, C. 2020. "Statistical Modelling and Stand Type Forest Mapping Selected Area around Istanbul Using Landsat-TM and Spot Data." *International Symposium on Remote Sensing and Integrated Technologies, 2002, 20-22 October, Istanbul, Turkey, Proceeding, 291-300.*

Appendix A Spectral signatures of MLC, RTC and SVM classifications, respectively

```
# Signatures Produced by ClassSig from
#    Class-Grid __1000001
#    and Stack __1000000

#    Number of selected grids
/*          1
#    Layer-Number    Band-name
/*          1       CIRimage_SegmentMeanShift_Cl.tif\Band_1

#  Type   Number of Classes   Number of Layers    Number of Parametric
Layers
    1             9                  1                         1
# ===================================================================

#  Class ID     Number of Cells    Class Name
        1             4527          bare
# Layers             1
# Means
             0.000000e+00
# Covariance
    1        0.000000e+00
# ------------------------------------------------------------------
#  Class ID     Number of Cells    Class Name
        2            25312          crops
# Layers             1
# Means
             9.053808e-01
# Covariance
    1        1.896503e+00
# -------------------------------------------------------------------
#  Class ID     Number of Cells    Class Name
        3             4612          pasture & scrub
# Layers             1
# Means
             2.252385e+00
# Covariance
    1        1.887278e-01
# -------------------------------------------------------------------
#  Class ID     Number of Cells    Class Name
        4            13605          planted
# Layers             1
# Means
             3.715619e+00
# Covariance
    1        2.035233e-01
# -------------------------------------------------------------------
#  Class ID     Number of Cells    Class Name
        5            11239          broadleaved
# Layers             1
```

```
# Means
            2.086663e+00
# Covariance
    1       3.992845e+00
# ---------------------------------------------------------------------
#  Class ID     Number of Cells    Class Name
        6            19363          larch
# Layers              1
# Means
            2.881888e+00
# Covariance
    1       6.097969e+00
# ---------------------------------------------------------------------
#  Class ID     Number of Cells    Class Name
        7            99145          mature evergreen
# Layers              1
# Means
            4.005245e+00
# Covariance
    1       1.529889e+00
# ---------------------------------------------------------------------
#  Class ID     Number of Cells    Class Name
        8            21437          young evergreen
# Layers              1
# Means
            7.000000e+00
# Covariance
    1       0.000000e+00
# ---------------------------------------------------------------------
#  Class ID     Number of Cells    Class Name
        9            18343          thick shrub land
# Layers              1
# Means
            4.888677e+00
# Covariance
    1       1.498292e+01
```

```
# Signatures Produced by ClassSig from
#    Class-Grid __1000001
#    and Stack __1000000

#    Number of selected grids
/*          1
#    Layer-Number    Band-name
/*          1       RTC_supervised.tif\Band_1

#  Type   Number of Classes   Number of Layers    Number of Parametric
Layers
    1             9                  1                         1
# ===================================================================

#  Class ID     Number of Cells    Class Name
       1              4527         bare
# Layers              1
# Means
             0.000000e+00
# Covariance
    1       0.000000e+00
# -------------------------------------------------------------------

#  Class ID     Number of Cells    Class Name
       2             25312         crops
# Layers              1
# Means
             1.949273e+00
# Covariance
    1       1.908928e+00
# -------------------------------------------------------------------

#  Class ID     Number of Cells    Class Name
       3              4612         pasture & scrub
# Layers              1
# Means
             2.544883e+00
# Covariance
    1       3.673193e-01
# -------------------------------------------------------------------

#  Class ID     Number of Cells    Class Name
       4             13605         planted
# Layers              1
# Means
             4.449394e+00
# Covariance
    1       2.247604e+00
# -------------------------------------------------------------------

#  Class ID     Number of Cells    Class Name
       5             11239         broadleaved
```

```
# Layers             1
# Means
             4.000000e+00
# Covariance
    1        0.000000e+00
# -------------------------------------------------------------------

#  Class ID     Number of Cells    Class Name
       6            19363           larch
# Layers             1
# Means
             5.094613e+00
# Covariance
    1        9.682203e-02
# -------------------------------------------------------------------

#  Class ID     Number of Cells    Class Name
       7            99145           mature evergreen
# Layers             1
# Means
             5.042977e+00
# Covariance
    1        1.958705e+00
# -------------------------------------------------------------------

#  Class ID     Number of Cells    Class Name
       8            21437           young evergreen
# Layers             1
# Means
             7.000000e+00
# Covariance
    1        0.000000e+00
# -------------------------------------------------------------------

#  Class ID     Number of Cells    Class Name
       9            18343           thick shrub land
# Layers             1
# Means
             7.921605e+00
# Covariance
    1        1.788937e-01
```

```
# Signatures Produced by ClassSig from
#    Class-Grid __1000001
#    and Stack __1000000

#    Number of selected grids
/*          1
#    Layer-Number    Band-name
/*          1       RTC_supervised.tif\Band_1

#  Type   Number of Classes   Number of Layers    Number of Parametric
Layers
   1            9                  1                         1
# ===================================================================

#  Class ID     Number of Cells    Class Name
       1              4527          bare
# Layers              1
# Means
             0.000000e+00
# Covariance
   1        0.000000e+00
# -------------------------------------------------------------------

#  Class ID     Number of Cells    Class Name
       2             25312          crops
# Layers              1
# Means
             1.949273e+00
# Covariance
   1        1.908928e+00
# -------------------------------------------------------------------

#  Class ID     Number of Cells    Class Name
       3              4612          pasture & scrub
# Layers              1
# Means
             2.544883e+00
# Covariance
   1        3.673193e-01
# -------------------------------------------------------------------

#  Class ID     Number of Cells    Class Name
       4             13605          planted
# Layers              1
# Means
             4.449394e+00
# Covariance
   1        2.247604e+00
# -------------------------------------------------------------------
```

```
#  Class ID     Number of Cells    Class Name
       5             11239          broadleaved
# Layers             1
# Means
             4.000000e+00
# Covariance
    1        0.000000e+00
# -------------------------------------------------------------------

#  Class ID     Number of Cells    Class Name
       6             19363          larch
# Layers             1
# Means
             5.094613e+00
# Covariance
    1        9.682203e-02
# -------------------------------------------------------------------

#  Class ID     Number of Cells    Class Name
       7             99145          mature evergreen
# Layers             1
# Means
             5.042977e+00
# Covariance
    1        1.958705e+00
# -------------------------------------------------------------------

#  Class ID     Number of Cells    Class Name
       8             21437          young evergreen
# Layers             1
# Means
             7.000000e+00
# Covariance
    1        0.000000e+00
# -------------------------------------------------------------------

#  Class ID     Number of Cells    Class Name
       9             18343          thick shrub land
# Layers             1
# Means
             7.921605e+00
# Covariance
    1        1.788937e-01
```

Appendix B: Error matrices for MLC< RTC, SVM classifications using ISO clusters as ground truth data

MLC_error_matrix2

OID	ClassValue	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_10	C_11	C_12	C_13	C_14	C_15	C_16	C_17	C_18	C_19	Total	U_Accuracy	Kappa
0	C_1	0	0	5	0	0	0	9	0	0	0	0	0	0	0	0	0	0	0	0	14	0	0
1	C_2	0	0	0	0	0	0	22	0	0	0	0	0	0	0	0	0	0	0	0	22	0	0
2	C_3	0	0	8	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	10	.800000	0
3	C_4	0	0	0	0	0	0	22	0	0	0	0	0	0	0	0	0	0	0	0	22	0	0
4	C_5	0	0	0	1	0	0	27	3	0	0	0	0	0	0	0	0	0	0	0	31	0	0
5	C_6	0	0	4	4	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	10	0	0
6	C_7	0	0	0	2	0	0	1	8	0	0	0	0	0	0	0	0	0	0	0	11	.090909	0
7	C_8	0	0	0	0	0	0	35	0	0	0	0	0	0	0	0	0	0	0	0	35	0	0
8	C_9	18	0	0	1	23	0	0	0	0	0	0	0	0	0	0	0	0	0	0	42	0	0
9	C_10	2	0	0	53	28	0	1	0	0	0	0	0	0	0	0	0	0	0	0	84	0	0
10	C_11	0	0	0	10	38	0	0	0	0	0	0	0	0	0	0	0	0	0	0	48	0	0
11	C_12	0	0	0	1	29	0	0	0	0	0	0	0	0	0	0	0	0	0	0	30	0	0
12	C_13	1	0	3	13	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	18	0	0
13	C_14	18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	18	0	0
14	C_15	20	0	0	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	22	0	0
15	C_16	5	0	4	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	12	0	0
16	C_17	5	0	10	12	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	27	0	0
17	C_18	32	0	0	4	0	7	1	0	0	0	0	0	0	0	0	0	0	0	0	44	0	0
18	C_19	6	0	3	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	10	0	0
19	Total	107	0	37	106	119	7	119	15	0	0	0	0	0	0	0	0	0	0	0	510	0	0
20	P_Accuracy	0	0	.216216	0	0	0	.008403	0	0	0	0	0	0	0	0	0	0	0	0	0	.017647	0
21	Kappa	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-.020786

RTC_error_matrix2

OID	ClassValue	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_10	C_11	C_12	C_13	C_14	C_15	C_16	C_17	C_18	C_19	Total	U_Accuracy	Kappa
0	C_1	0	0	1	0	0	0	13	0	0	0	0	0	0	0	0	0	0	0	0	14	0	0
1	C_2	0	0	0	0	0	0	22	0	0	0	0	0	0	0	0	0	0	0	0	22	0	0
2	C_3	0	0	0	0	0	0	3	7	0	0	0	0	0	0	0	0	0	0	0	10	0	0
3	C_4	0	0	0	0	0	0	21	1	0	0	0	0	0	0	0	0	0	0	0	22	0	0
4	C_5	0	0	0	0	0	0	5	26	0	0	0	0	0	0	0	0	0	0	0	31	0	0
5	C_6	0	0	0	5	0	0	0	5	0	0	0	0	0	0	0	0	0	0	0	10	0	0
6	C_7	0	0	0	0	0	0	3	8	0	0	0	0	0	0	0	0	0	0	0	11	.272727	0
7	C_8	0	0	0	0	0	0	24	11	0	0	0	0	0	0	0	0	0	0	0	35	.314286	0
8	C_9	0	0	0	0	0	0	42	0	0	0	0	0	0	0	0	0	0	0	0	42	0	0
9	C_10	0	0	0	50	0	0	34	0	0	0	0	0	0	0	0	0	0	0	0	84	0	0
10	C_11	0	0	0	48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	48	0	0
11	C_12	0	0	0	10	19	0	1	0	0	0	0	0	0	0	0	0	0	0	0	30	0	0
12	C_13	0	1	2	14	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	18	0	0
13	C_14	0	1	0	0	0	0	5	0	12	0	0	0	0	0	0	0	0	0	0	18	0	0
14	C_15	0	0	0	0	14	0	8	0	0	0	0	0	0	0	0	0	0	0	0	22	0	0
15	C_16	9	0	2	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	12	0	0
16	C_17	0	0	19	1	7	0	0	0	0	0	0	0	0	0	0	0	0	0	0	27	0	0
17	C_18	0	13	0	0	7	23	0	0	1	0	0	0	0	0	0	0	0	0	0	44	0	0
18	C_19	3	0	7	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	10	0	0
19	Total	12	15	31	128	48	23	181	59	13	0	0	0	0	0	0	0	0	0	0	510	0	0
20	P_Accuracy	0	0	0	0	0	0	.016575	.186441	0	0	0	0	0	0	0	0	0	0	0	0	.027451	0
21	Kappa	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-.011209

SVM_error_matrix2

OID	ClassValue	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_10	C_11	C_12	C_13	C_14	C_15	C_16	C_17	C_18	C_19	Total	U_Accuracy	Kappa
0	C_1	0	0	10	0	0	0	4	0	0	0	0	0	0	0	0	0	0	0	0	14	0	0
1	C_2	0	0	0	0	0	0	22	0	0	0	0	0	0	0	0	0	0	0	0	22	0	0
2	C_3	0	0	1	0	0	0	1	8	0	0	0	0	0	0	0	0	0	0	0	10	.100000	0
3	C_4	0	0	0	0	0	0	17	5	0	0	0	0	0	0	0	0	0	0	0	22	0	0
4	C_5	0	0	0	0	0	0	0	31	0	0	0	0	0	0	0	0	0	0	0	31	0	0
5	C_6	0	0	0	2	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	10	0	0
6	C_7	0	0	0	0	0	0	11	0	0	0	0	0	0	0	0	0	0	0	0	11	1	0
7	C_8	0	0	0	0	0	0	35	0	0	0	0	0	0	0	0	0	0	0	0	35	0	0
8	C_9	0	0	0	0	0	0	41	0	1	0	0	0	0	0	0	0	0	0	0	42	.023810	0
9	C_10	0	0	0	0	0	0	84	0	0	0	0	0	0	0	0	0	0	0	0	84	0	0
10	C_11	0	0	0	3	0	0	45	0	0	0	0	0	0	0	0	0	0	0	0	48	0	0
11	C_12	0	0	0	0	0	0	30	0	0	0	0	0	0	0	0	0	0	0	0	30	0	0
12	C_13	0	1	4	12	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	18	0	0
13	C_14	0	0	0	0	0	2	0	0	16	0	0	0	0	0	0	0	0	0	0	18	0	0
14	C_15	0	0	0	0	0	0	22	0	0	0	0	0	0	0	0	0	0	0	0	22	0	0
15	C_16	9	0	0	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	12	0	0
16	C_17	0	1	17	7	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	27	0	0
17	C_18	0	21	0	0	6	12	4	0	1	0	0	0	0	0	0	0	0	0	0	44	0	0
18	C_19	6	0	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	10	0	0
19	Total	15	23	36	27	9	14	316	52	18	0	0	0	0	0	0	0	0	0	0	510	0	0
20	P_Accuracy	0	0	.027778	0	0	0	.034810	0	.055556	0	0	0	0	0	0	0	0	0	0	0	.025490	0
21	Kappa	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-.005997

Append C Spectral signatures of 9 training sample classes according to isodata classification of images.

planted

Statistics	Band_1	Band_2	Band_3
Minimum	33.00	32.00	24.00
Maximum	197.00	230.00	192.00
Mean	116.57	109.34	90.68
Std.dev	24.73	26.82	22.46
Covariance			
Band_1	611.55	520.82	500.69
Band_2	520.82	719.11	574.86
Band_3	500.69	574.86	504.67

broadlea...

Statistics	Band_1	Band_2	Band_3
Minimum	44.00	46.00	37.00
Maximum	208.00	195.00	172.00
Mean	140.02	116.64	100.37
Std.dev	28.64	25.71	23.28
Covariance			
Band_1	819.98	618.32	592.30
Band_2	618.32	660.78	583.79
Band_3	592.30	583.79	542.16

larch

Statistics	Band_1	Band_2	Band_3
Minimum	46.00	38.00	36.00
Maximum	220.00	192.00	171.00
Mean	137.53	105.32	96.58
Std.dev	33.02	28.29	27.07
Covariance			
Band_1	1090.19	892.09	862.57
Band_2	892.09	800.45	759.40
Band_3	862.57	759.40	732.75

mature e...

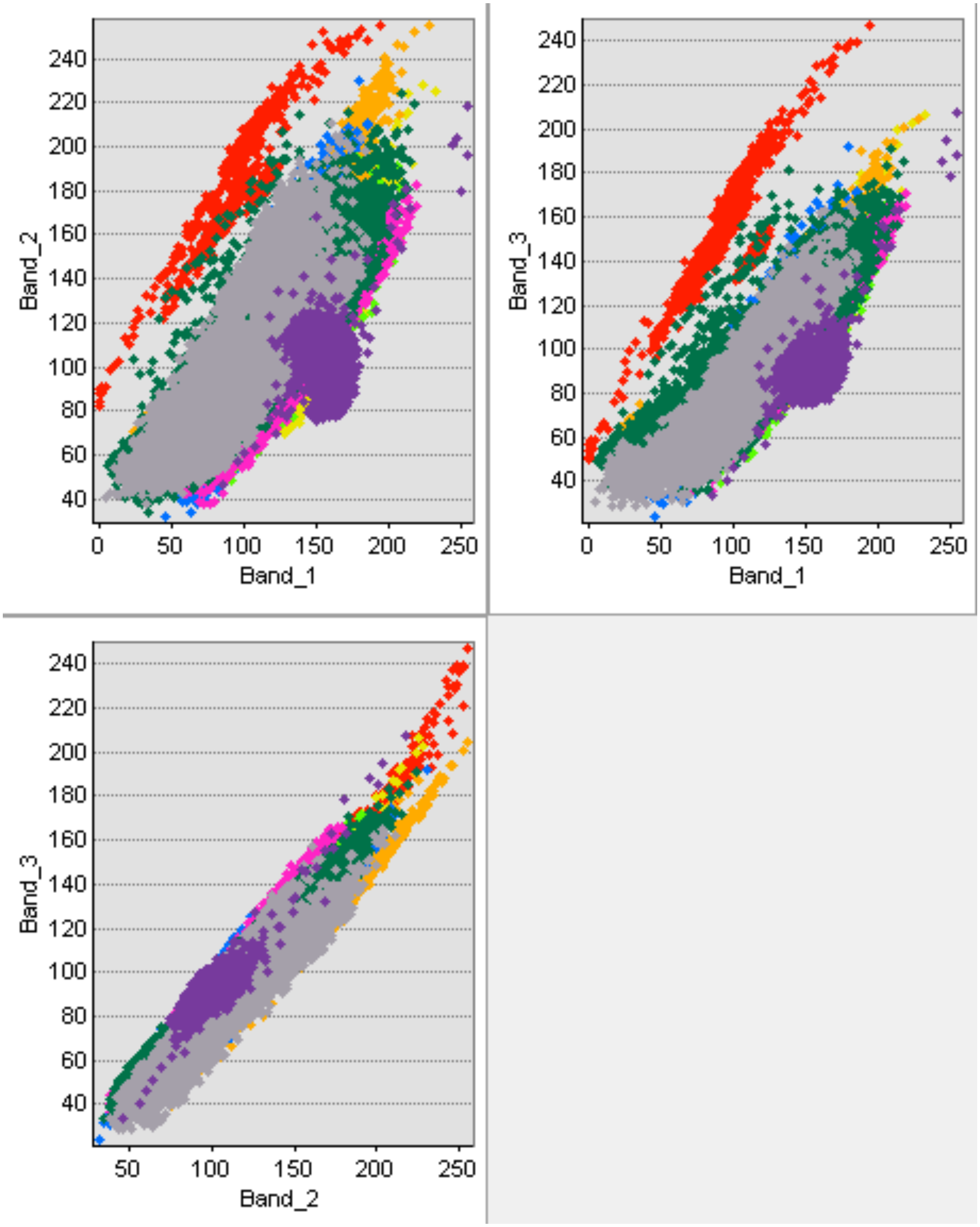
Statistics	Band_1	Band_2	Band_3
Minimum	6.00	34.00	34.00
Maximum	218.00	224.00	191.00
Mean	103.37	95.91	81.94
Std.dev	36.15	28.96	24.07
Covariance			
Band_1	1306.52	978.85	804.56
Band_2	978.85	838.95	684.71
Band_3	804.56	684.71	579.34

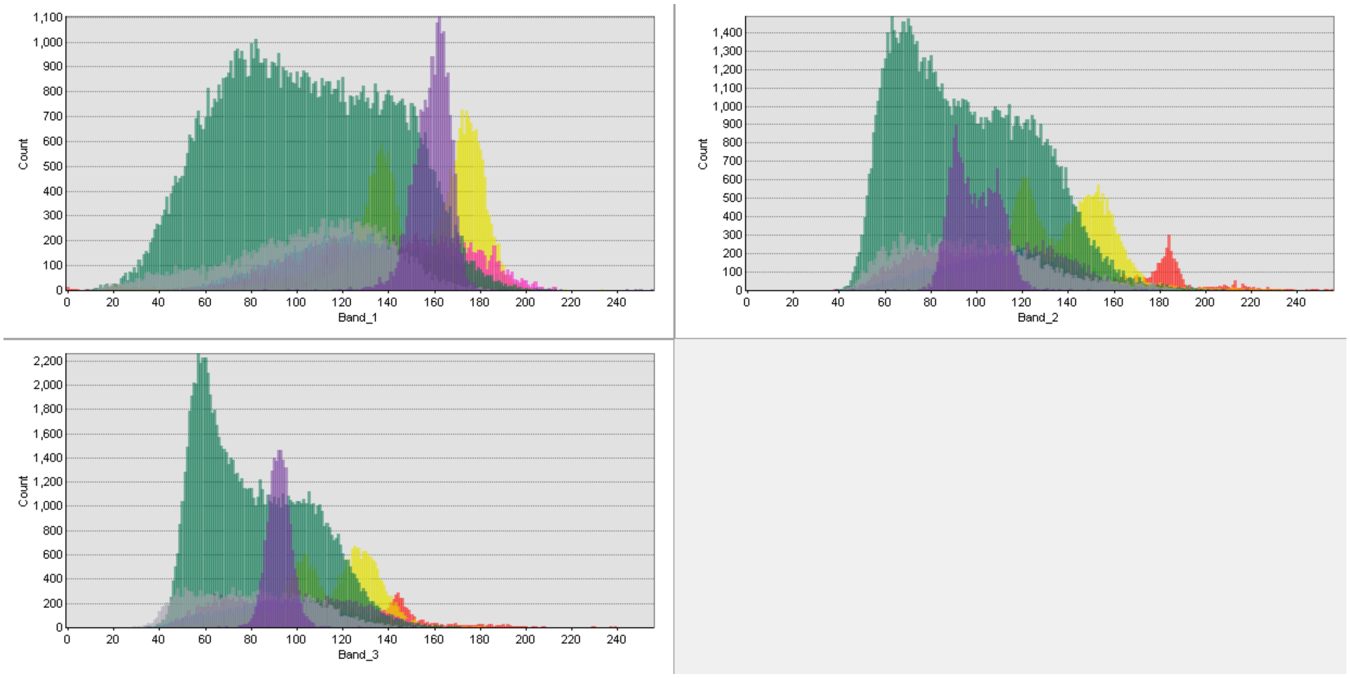
young e...

Statistics	Band_1	Band_2	Band_3
Minimum	5.00	38.00	29.00
Maximum	186.00	211.00	165.00
Mean	104.84	98.22	80.26
Std.dev	33.26	30.07	24.65
Covariance			
Band_1	1106.49	779.25	729.46
Band_2	779.25	904.25	702.27
Band_3	729.46	702.27	607.63

thick shr...

Statistics	Band_1	Band_2	Band_3
Minimum	86.00	46.00	34.00
Maximum	255.00	218.00	207.00
Mean	159.09	99.42	92.19
Std.dev	8.34	10.49	6.40
Covariance			
Band_1	69.51	-18.68	18.50
Band_2	-18.68	110.14	52.26
Band_3	18.50	52.26	40.98


Band_2
Band_1
Band_3
Band_1
Band_3
Band_2


Count
Band_1
Count
Band_2
Count
Band_3

Question 1-2 Biomass estimation based on simple random sampling

Estimates of above-ground biomass across the site were first generated using a regression analysis of 50 randomly sampled field plots and linked LiDAR data points in exercise 03. From 8 regression analyses, two candidate regression models were selected from previous exercise (Table 1). However, since image classifications and LiDAR estimates were based on the full dataset of 50 observations, the second candidate model was omitted. Therefore, parameters from the nonlinear least squares regression model (M4) were fitted to the biomass formula below (1).

$$7.03 * \text{“MEANH”} + 146.78 * \text{“COVER”} + 12.72 * \text{“SD”} \quad (1)$$

Table 1 Regression models and parameters used for calculating above ground-biomass. AIC: Akaike Criterion Information, BIC: Bayesian Information Criterion.

	RMSE(%)	AIC	BIC	Theil's U	Paramter estimates (SE)		
					MeanH	SD	Cover
M4.NLS.comp	16.30(7.4)	427.77	439.24	0.998	7.03(2.16)**	12.72(2.16)***	146.78(51.59)**
M5.NLS.clean	15.18(6.8)	404.08	415.31	0.998	6.95(2.27)**	13.23(2.82)***	159.11(91.51)

Based on simple random sampling, mean biomass density was estimated as 220.63Mt/ha (95% CI 195.64, 245.62). Confidence intervals were also generated using the *DescTools* package in R studio (95% CI 195.01, 246.25). As *DescTools* estimates presented a higher certainty than the first estimates, the manually generated confidence intervals were selected for computing the following estimations.

Calculating the sample area from pixel dimensions in the 'property>source' tab in ArcGis, the total area of the site was 107.75ha. Using this value, the total sum of biomass was equal to 23,772.76Mt (95% CI 21,080.21, 26,465.08). Based on market price of £50/Mt, total forest biomass was worth £1,118,638 (95% CI 1,054,010.5, 1,323,254).

```
> MeanCI(MLC_stratified_biomass$BIOMASS, conf.level = 0.95)
    mean   lwr.ci   upr.ci
220.6280 195.0084 246.2476
> stdev <- sd(MLC_stratified_biomass$BIOMASS)
> stderr <- sqrt(stdev^2/50)
> 220.628 - 1.96 * stderr
[1] 195.6404
> 220.628 + 1.96 * stderr
[1] 245.6156
```

Figure 1. Biomass density & 95% CI (*DescTools* and Gaussian equations)

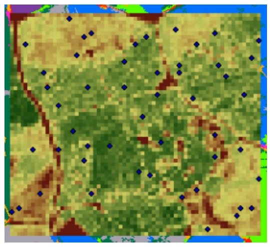


Figure 2. LiDAR estimate (M4)

Question 3-4 Biomass estimation based on remote imagery classification.

Based on image classification of remote imagery, which was generated using the maximum likelihood algorithm, weighted statistics of biomass were calculated for the forest classes (Table 2). Post-stratification of classes was conducted using a weighting system based on the sampling derivative of the proportion of class areas to the total area of forested land. The total area of forested land was 97.8ha. The largest forest class was 'Evergreen' (83.4ha), which accounted for 85.2% of the forested area, followed by 'Mixed Broadleaved' (7.1%, 6.9ha), 'Larch' (6.1%, 6.0ha), and 'Planted' (1.6%, 1.6ha). These calculations are presented below (Figure 2).

Weighted estimates were generated twice (Figure 2-3), first using the *Hmisc* package in Rstudio (225.07Mt/ha, 95% CI 50.71, 399.43), and second using manual calculations (226.40Mt/ha, 95% CI 221.52, 231.29). Due to higher uncertainty generated by *Hmisc* package, the manually generated results were used in the following estimations. Based on weighted estimates, the total sum of biomass was equal to 21,141.92Mt (95% CI 21.664.66, 22,620.16).

Table 2. Estimates of biomass among forest classes using random sampling and stratified sampling.

	n	Mean	SD	95% CI		Tot Vol	Tot Val	Tukey Contrasts of Means (x/y)				
		(Mt/ha)		Lower	Upper	(Mt)	(£)	C4	C5	C6	C7	
1. C4 Planted	1	159.30	/	/	/	248.8	12,440	-	44.72	-49.95	-78.90	**
2. C5 Mix_Broad	6	114.58	22.88	90.38	138.60	794.9	37,495	-44.72	-	-94.67	-123.61	
3. C6 Larch	2	209.25	40.66	156.05	574.55	1,253.4	62,670	49.95	94.67	-	-28.95	
4. C7 Evergreen	41	**238.20**	88.00	210.42	265.98	19,854.8	979 ,200	78.90	123.61	28.95	-	
5. All_Field_Plots	50	220.63	90.16	195.01	246.25	21,577.4	1,078,870					
6. All_Image_Classes	50	226.40	88.96	221.52	231.29	22,141.9	1,107,095					

```
> #calculate weights for each class
> classest.forest$AREA[1]/10000/area_ha_forest_class
[1] 0.01596023
> classest.forest$AREA[2]/10000/area_ha_forest_class
[1] 0.07090529
> classest.forest$AREA[3]/10000/area_ha_forest_class
[1] 0.06122449
> classest.forest$AREA[4]/10000/area_ha_forest_class
[1] 0.85191
```

Figure 2. Class weights generated using R-studio

```
wtd.mean(MLC_stratified_biomass$BIOMASS,
         weights = MLC_stratified_biomass$RASTERVALU)
] 225.0703
wtd.mean(MLC_stratified_biomass$BIOMASS,
                    weights = MLC_stratified_biomass$RASTERVA
] 225.0703
wtd.var(MLC_stratified_biomass$BIOMASS,
                      weights = MLC_stratified_biomass$RASTER
] 7914.023
weighted_mean + 1.96*weighted_sd
] 399.4334
weighted_mean - 1.96*weighted_sd
] 50.70711
```

```
(mean(Planted$BIOMASS))*weight_class4 +
               (mean(Mixed_Broadleaved$BIOMASS))*weight_cla
               (mean(Larch$BIOMASS))*weight_class6 +
               (mean(Evergreen$BIOMASS))*weight_class7
] 226.4011
weight_class4^2 * (sd.4^2)/n.4 +
                    weight_class5^2 * (sd.5^2)/n.5 +
                    weight_class6^2 * (sd.6^2)/n.6 +
                    weight_class7^2 * (sd.7^2)/n.7 -
                    weight_class4 * (sd.4^2)/N -
                    weight_class5 * (sd.5^2)/N -
                    weight_class6 * (sd.6^2)/N -
                    weight_class7 * (sd.7^2)/N
] 5.905392
sqrt(weighted.variance)
] 2.430101
total.w.mean - qt(1 - 0.05/2, 50-1) * weighted.stderr
] 221.5177
total.w.mean + qt(1 - 0.05/2, 50-1) * weighted.stderr
] 231.2846
```

Figure 3. Bottom left: Stratified estimates of biomass generated manually using Gaussian formula and image classification. Right: Stratified estimates of biomass generated using *Hmisc* package and image classifications.

Question 5. Profit comparison between unweighted & weighted biomass estimates using survey costs

Comparing between the aggregated results above ("5. All - Field Plots", "6. All – Image Classification"), we find that higher values were generated from the stratified sampling of optical imagery than from the unweighted random sampling of field plots. Applying the principle of conservativeness (Grassi et al. 2008; Knoke 2013), profits were compared using lower bounds of estimates set at the level of 95% confidence. Based on conservative estimates of the unweighted random sampling, biomass sales were worth £1,054,010.5. Based on conservative estimates of the stratified sampling of optical imagery, biomass sales were worth £1,083,232.8. Therefore, using image classification methods and a camera survey provided an increase in profits of £29,222.3.

Considering the cost of the camera survey was £2 per hectare and the total area of forest was 97.8ha, the LiDAR survey required a total budget of £195.69. Therefore, the value of investing in image classification methods was worth £29,026.61. This estimate was much lower than the figure presented in the exercise video. This difference may result from the varying regression parameters fitted to the LiDAR biomass estimation (Table 1). However, this appears somewhat unlikely because the difference between these figures here and those in the video is so significant.

Question 6-9. Profit comparison between all three biomass estimations using survey costs

Using image classifications and weighting system as above (proportionate to total forest area), biomass estimation was carried our using LiDAR datapoints for the same four forest classes (Table 3). These LiDAR values were generated using 'zonal statistics' in the 'spatial analyst' toolbox of ArcGis software. Using similar formula of variance, standard error and Gaussian distribution, this produced a mean biomass density of 247.55Mt/ha and lower and upper bounds of 247.17 and 247.93, respectively. From this, the total forest biomass was estimated to be 24,210.39Mt (95% CI 24,173.23, 24,247.55).

Table 3	*n*	Mean (Mt/ha)	SD	95% CI Lower	95% CI Upper	Tot Vol (Mt)	Tot Val (£)
5. All_Field_Plots	50	220.63	90.16	195.01	246.25	21,577.4	1,078,870
6. All_Image_Classes	50	226.40	88.96	221.52	231.29	22,141.9	1,107,095
7. All_LiDAR_Im_C	50	247.55	0.19	247.17	247.93	24,210.4	1,210,520

Comparing between the aggregated results above ("5. All - Field Plots", "6. All – Image Classification", "7. All – LiDAR & Image Classes"), we find that values generated from the LiDAR data were higher than from other methods.

With all estimates now available, results confirmed a lack of bias as no methods produced a mean beyond the range of confidence intervals of other methods. Applying the principle of conservativeness, biomass sales were compared using lower bounds of estimates set at the level of 95% confidence. Accordingly, conservative estimates generated using LiDAR data points and weighted stratification from the image classification (7.All – LiDAR & Image Classification), total biomass sales were worth £1,208,661.3. Therefore, the LiDAR survey provided an increase in estimated sales of £125,428.5 compared to stratified image classification. Compared to the initial, simple random sampling of field plots, LiDAR estimates increased even further by £154,650.8.

If we take into account the extra cost of LiDAR surveying, we can evaluate the value of investing in this method. If the cost of LiDAR survey was £10 per hectare and the total area of forest was 97.8ha, the LiDAR survey required a total budget of £970.80. Compared to image classification, LiDAR surveying provided the added value of £124,457.7. Compared to simple random sampling of field plots, LiDAR survey provided the added value of £153,680.

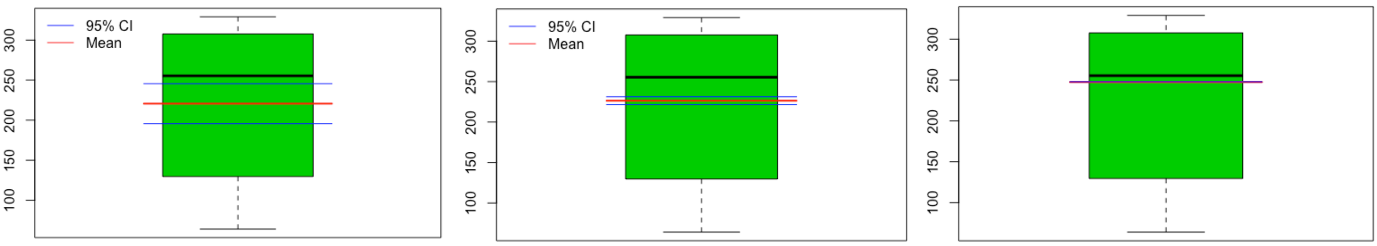


Figure 4. Boxplot graph comparing means and confidence intervals from the three estimation methods. Left: Simple random sampling and unweighted estimations. Centre: Image classification and weighted estimations. Right: LiDAR estimations using weighted classification from optical imagery.

Q 10. Future recommendations

In order to provide a comprehensive cost-efficiency analysis of remote sensing, it might be worthwhile to compare between different image classification methods, particularly if the forested site was a significantly large area. Accuracy assessments in exercise 2 indicated that the Support Vector Machine algorithm produced higher levels of accuracy than the Maximum Likelihood algorithm. However, no significant differences were observed between these methods, which were likely the result of the small land areas of the sample site and the four forest classes.

In exercise 03, regression analyses were assessed before fitting them to LiDAR biomass estimation models. It might be useful for statistics newcomers like myself to follow a recommended workflow of regression diagnostics? In addition, since some tests showed certain violations more significantly than others, it seems that such model assessment toolkit or protocol would be highly beneficial. As a suggestion, would ether the Bonferroni correction test or the Holm-Bonferroni step-by-step methods seem relevant here? In particular, there a number of common pitfalls to look out for such as 1) multi-collinearity and 2) influence and leverage of outliers. Both were identified in excerise03.

Regarding the former, studies have shown that collinear variables can give the appearance of a highly significant model (Graham 2003; Sawadogo et al. 2010), which can lead to over-estimates of biomass (Basuki et al. 2009; Colgan, Asner, and Swemmer 2013). To avoid such errors, some have recommended the application of variance inflation factor metrics (Chave et al. 2005; Sileshi 2014, 249), or the use of step-wise regression methods (Carrascal, Galván, and Gordo 2009; Ruben Valbuena et al. 2012).

Regarding the latter, the potential effect of outliers has led some to recommend use of robust regression (Sileshi 2014, 248). I am unsure whether such methods apply to the current project. If so, a number of estimators need exploring, such as the M-estimators and S-estimators. Results from exercise03 indicated that Tukey’s robust regression produced lowest RSME with the current dataset, though evidence of over-fitting was also indicated. In order to choose between candidate models, it is unclear which information criteria and which cross validation methods to rely given the nature of the data and models. Some recommend AIC and BIC scoring (Lukacs, Burnham, and Anderson 2010), others such these methods assume an optimal number of predictors are in use (Guthery et al. 2005). Alternatively, others recommend the use of Theil’s uncertainty scoring (Valbuena et al. 2017). This seems to be an emerging area of research, nonetheless an important aspect of remote sensing estimations of biomass and a worthwhile protocol for developing and adopting in this workflow.

Bibliography

Basuki, T M, P E Van Laake, A K Skidmore, and Y A Hussin. 2009. “Allometric Equations for Estimating the Above-Ground Biomass in Tropical Lowland Dipterocarp Forests.” *Forest ecology and management* 257(8): 1684–94.

Carrascal, Luis M, Ismael Galván, and Oscar Gordo. 2009. “Partial Least Squares Regression as an Alternative to Current Regression Methods Used in Ecology.” *Oikos* 118(5): 681–90.

Chave, Jérôme et al. 2005. “Tree Allometry and Improved Estimation of Carbon Stocks and Balance in Tropical Forests.” *Oecologia* 145(1): 87–99.

Colgan, Matthew S, Gregory P Asner, and Tony Swemmer. 2013. “Harvesting Tree Biomass at the Stand Level to Assess the Accuracy of Field and Airborne Biomass Estimation in Savannas.” *Ecological Applications* 23(5): 1170–84.

Graham, Michael H. 2003. “Confronting Multicollinearity in Ecological Multiple Regression.” *Ecology* 84(11): 2809–15.

Grassi, Giacomo et al. 2008. “Applying the Conservativeness Principle to REDD to Deal with the Uncertainties of the Estimates.” *Environmental Research Letters* 3(3): 35005.

Guthery, Fred S, Leonard A Brennan, Markus J Peterson, and Jeffrey J Lusk. 2005. “Information Theory in Wildlife Science: Critique and Viewpoint.” *The Journal of Wildlife Management* 69(2): 457–65.

Knoke, Thomas. 2013. “Uncertainties and REDD+: Implications of Applying the Conservativeness Principle to Carbon Stock Estimates.” *Climatic Change* 119(2): 261–67.

Lukacs, Paul M, Kenneth P Burnham, and David R Anderson. 2010. “Model Selection Bias and Freedman’s Paradox.” *Annals of the Institute of Statistical Mathematics* 62(1): 117.

Sawadogo, Louis et al. 2010. “Allometric Prediction of Above-Ground Biomass of Eleven Woody Tree Species in the Sudanian Savanna-Woodland of West Africa.” *Journal of Forestry Research* 21(4): 475–81.

Sileshi, Gudeta W. 2014. “A Critical Review of Forest Biomass Estimation Models, Common Mistakes and Corrective Measures.” *Forest Ecology and Management* 329: 237–54.

Valbuena, R et al. 2017. “Enhancing of Accuracy Assessment for Forest Above-Ground Biomass Estimates Obtained from Remote Sensing via Hypothesis Testing and Overfitting Evaluation.” *Ecological modelling* 366: 15–26.

Valbuena, Ruben, Francisco Mauro, Roberto Rodriguez-Solano, and Jose Antonio Manzanera. 2012. “Partial Least Squares for Discriminating Variance Components in Global Navigation Satellite Systems Accuracy Obtained under Scots Pine Canopies.” *Forest Science* 58(2): 139–53.

R Syntax:

```
### MSc Inventory Monitoring and Assessment ##
###  Assignment 4; Cost-efficieny of methods ####
############ S.Murphy 02-05-2020 ##########

# using subset function
sapply(MLC_stratified_biomass, class)
MLC_stratified_biomass$RASTERVALU_factor <- factor(MLC_stratified_biomass$RASTERVALU)
MLC_stratified_biomass$RASTERVALU_factor
Planted <- subset(MLC_stratified_biomass, RASTERVALU_factor=="4")
Mixed_Broadleaved <- subset(MLC_stratified_biomass, RASTERVALU_factor=="5")
Larch <- subset(MLC_stratified_biomass, RASTERVALU_factor=="6")
Evergreen <- subset(MLC_stratified_biomass, RASTERVALU_factor=="7")
```

```
#prices
price.wood <- 50
price.camera <- 2
price.lidar <- 10

# total area within area outlined by CIR image in Ex03: 107.7504 hectares
area_ha <- 61 * 69 * 16 * 16 / 10000

# simple random sampling
mean_biomass <- mean(ClocaenogField$BIOMASS)
describe(mean_biomass)

total <- mean_biomass * area_ha
total * price.wood

library(readxl)
MLC_stratified_biomass_sum <- read_excel("Google Drive/Bangor /Assessments/Inv Assessment & Monitoring/Ex04_CostEfficiencyof_InvMethods/Ex04_Output/MLC_stratified_biomass_sum.xls")
View(MLC_stratified_biomass_sum)
classest.forest <- MLC_stratified_biomass_sum[4:7,]
View(classest.forest)
sum(classest.forest$AREA)/10000

# total area of forested classes: 97.8432 hectares
area_ha_forest_class <- sum(classest.forest$AREA)/10000

# calculate total biomass of forest classes
total_biomass_forest_class <- mean_biomass * area_ha_forest_class
total_biomass_forest_class

#calculate weights for each class
weight_class4 <- classest.forest$AREA[1]/10000/area_ha_forest_class
weight_class5 <- classest.forest$AREA[2]/10000/area_ha_forest_class
weight_class6 <- classest.forest$AREA[3]/10000/area_ha_forest_class
weight_class7 <- classest.forest$AREA[4]/10000/area_ha_forest_class


#stratified samples
N <- length(MLC_stratified_biomass$BIOMASS)

#strata estimates
stratum4 <- MLC_stratified_biomass[MLC_stratified_biomass$RASTERVALU==4,]
mean.4 <- mean(stratum4$BIOMASS)
sd.4 <- 0
n.4 <- length(stratum4$BIOMASS)

stratum5 <- MLC_stratified_biomass[MLC_stratified_biomass$RASTERVALU==5,]
mean.5 <- mean(stratum5$BIOMASS)
sd.5 <- sd(stratum5$BIOMASS)
n.5 <- length(stratum5$BIOMASS)

stratum6 <- MLC_stratified_biomass[MLC_stratified_biomass$RASTERVALU==6,]
mean.6 <- mean(stratum6$BIOMASS)
sd.6 <- sd(stratum6$BIOMASS)
n.6 <- length(stratum6$BIOMASS)

stratum7 <- MLC_stratified_biomass[MLC_stratified_biomass$RASTERVALU==7,]
mean.7 <- mean(stratum7$BIOMASS)
sd.7 <- sd(stratum7$BIOMASS)
n.7 <- length(stratum7$BIOMASS)
```

```
#weighted mean
total.w.mean <- (mean(Planted$BIOMASS))*weight_class4 +
            (mean(Mixed_Broadleaved$BIOMASS))*weight_class5 +
            (mean(Larch$BIOMASS))*weight_class6 +
            (mean(Evergreen$BIOMASS))*weight_class7

total.w.biomass <- total.w.mean*area_ha_forest_class
total.w.price <- total.w.biomass*price.wood

#lower confidence intervals
weighted.variance <- weight_class4^2 * (sd.4^2)/n.4 +
                weight_class5^2 * (sd.5^2)/n.5 +
                weight_class6^2 * (sd.6^2)/n.6 +
                weight_class7^2 * (sd.7^2)/n.7 -
                weight_class4 * (sd.4^2)/N -
                weight_class5 * (sd.5^2)/N -
                weight_class6 * (sd.6^2)/N -
                weight_class7 * (sd.7^2)/N
(weighted.variance * N)

weighted.stderr <- sqrt(weighted.variance)
weighted.low.ci <- total.w.mean - qt(1 - 0.05/2, 50-1) * weighted.stderr
weighted.high.ci <- total.w.mean + qt(1 - 0.05/2, 50-1) * weighted.stderr

library(Hmisc)
weighted_mean <- wtd.mean(MLC_stratified_biomass$BIOMASS,
               weights = MLC_stratified_biomass$RASTERVALU)
weighted_var <- wtd.var(MLC_stratified_biomass$BIOMASS,
                 weights = MLC_stratified_biomass$RASTERVALU)
weighted_sd <- sqrt(weighted_var)
weighted_CI_upper <- weighted_mean + 1.96*weighted_sd
weighted_CI_lower <-

wtd.mean(Planted$BIOMASS, weights = Planted$RASTERVALU, na.rm = TRUE)
wtd.mean(Mixed_Broadleaved$BIOMASS, weights = Mixed_Broadleaved$RASTERVALU, na.rm = TRUE)
wtd.mean(Larch$BIOMASS, weights = Larch$RASTERVALU, na.rm = TRUE)
wtd.mean(Evergreen$BIOMASS, weights = Evergreen$RASTERVALU, na.rm = TRUE)

wtd.var(Planted$BIOMASS, weights = Planted$RASTERVALU, na.rm = TRUE)
wtd.var(Mixed_Broadleaved$BIOMASS, weights = Mixed_Broadleaved$RASTERVALU, na.rm = TRUE)
wtd.var(Larch$BIOMASS, weights = Larch$RASTERVALU, na.rm = TRUE)
wtd.var(Evergreen$BIOMASS, weights = Evergreen$RASTERVALU, na.rm = TRUE)
weighted.var(MLC_stratified_biomass$BIOMASS, weights = MLC_stratified_biomass$RASTERVALU, na.rm = TRUE)



#lidar estimation
lidar.weighted.mean <- sum(classest.forest$MEAN * classest.forest$AREA/10000/area_ha_forest_class)
total.w.lidar <- lidar.weighted.mean * area_ha_forest_class
total.lidar.price <- total.w.lidar * price.wood
#variance
lidar.variance.dummy <- sum(classest.forest$STD^2 * (classest.forest$AREA/10000/area_ha_forest_class)^2)
#residual variance
lidar.resid.variance <- (1-50/sum(classest.forest$COUNT) *
                  sum((MLC_stratified_biomass$BIOMASS) -
                     7.03 * (MLC_stratified_biomass$MEANH) +
                     146.78 * (MLC_stratified_biomass$COVER) +
                     12.72 * (MLC_stratified_biomass$SD))^2 /
                  (50*(50-3-1)))

lidar.variance <- lidar.variance.dummy + lidar.resid.variance
stderr.lidar <- sqrt(lidar.variance/ sum(classest.forest$COUNT))
```

```
lidar.low.ci <- lidar.weighted.mean - qt(1 - 0.05/2, 50-1) * stderr.lidar
lidar.high.ci <- lidar.weighted.mean  + qt(1 - 0.05/2, 50-1) * stderr.lidar



mean.f.p <- mean_biomass
ci.low.f.p <- ci.low.fp.dummy
ci.high.f.p <-ci.high.fp.dummy

mean.i.c <- total.w.mean
ci.low.i.c <- weighted.low.ci
ci.high.i.c <- weighted.high.ci

mean.lid <- lidar.weighted.mean
ci.low.lid <- lidar.low.ci
ci.high.lid <- lidar.high.ci


ggplot(MLC_stratified_biomass, aes(RASTERVALU_factor,BIOMASS)) + geom_boxplot()
boxplot(MLC_stratified_biomass$BIOMASS, col = 3)
lines(c(0.75, 1.25),c(ci.low.f.p, ci.low.f.p),col=4)
lines(c(0.75, 1.25),c(mean.f.p, mean.f.p),col=2,lwd=2)
lines(c(0.75,1.25),c(ci.high.f.p, ci.high.f.p), col=4)
legend("topleft", c("95% CI", "Mean"), lty=1,col = c(4, 2),bty ="n")

ggplot(MLC_stratified_biomass, aes(RASTERVALU_factor,BIOMASS)) + geom_boxplot()
boxplot(MLC_stratified_biomass$BIOMASS, col = 3)
lines(c(0.75, 1.25),c(ci.low.i.c, ci.low.i.c),col=4)
lines(c(0.75, 1.25),c(mean.i.c, mean.i.c),col=2,lwd=2)
lines(c(0.75,1.25),c(ci.high.i.c, ci.high.i.c), col=4)
legend("topleft", c("95% CI", "Mean"), lty=1,col = c(4, 2),bty ="n")

ggplot(MLC_stratified_biomass, aes(RASTERVALU_factor,BIOMASS)) + geom_boxplot()
boxplot(MLC_stratified_biomass$BIOMASS, col = 3)
lines(c(0.75, 1.25),c(ci.low.lid, ci.low.lid),col=4)
lines(c(0.75, 1.25),c(mean.lid, mean.lid),col=2,lwd=2)
lines(c(0.75,1.25),c(ci.high.lid, ci.high.lid), col=4)
legend("topleft", c("95% CI", "Mean"), lty=1,col = c(4, 2),bty ="n")


mean_biomass <- mean(MLC_stratified_biomass$BIOMASS)
stdev <- sd(MLC_stratified_biomass$BIOMASS)
stderr <- sqrt(stdev^2 / 50 )
ci.low.fp.dummy <- mean_biomass - 1.96 * stderr
ci.high.fp.dummy <- mean_biomass + 1.96 * stderr

weighted.stderr <- sqrt(weighted.variance)
weighted.low.ci <- total.w.mean - qt(1 - 0.05/2, 50-1) * weighted.stderr
weighted.high.ci <- total.w.mean + qt(1 - 0.05/2, 50-1) * weighted.stderr


total.w.mean <- (mean(Planted$BIOMASS))*weight_class4 +
  (mean(Mixed_Broadleaved$BIOMASS))*weight_class5 +
  (mean(Larch$BIOMASS))*weight_class6 +
  (mean(Evergreen$BIOMASS))*weight_class7
weighted.stderr <- sqrt(weighted.variance)
weighted.low.ci <- total.w.mean - qt(1 - 0.05/2, 50-1) * weighted.stderr
weighted.high.ci <- total.w.mean + qt(1 - 0.05/2, 50-1) * weighted.stderr


lidar.weighted.mean <- sum(classest.forest$MEAN * classest.forest$AREA/10000/area_ha_forest_class)
stderr.lidar <- sqrt(lidar.variance/ sum(classest.forest$COUNT))
```

```
lidar.low.ci <- lidar.weighted.mean - qt(1 - 0.05/2, 50-1) * stderr.lidar
lidar.high.ci <- lidar.weighted.mean  + qt(1 - 0.05/2, 50-1) * stderr.lidar




#confidence intervals:
library(DescTools)
MeanCI(ClocaenogField$BIOMASS, conf.level = 0.95, na.rm=TRUE)
MeanCI(MLC_stratified_biomass$BIOMASS, conf.level = 0.95)
describe(MLC_stratified_biomass$BIOMASS)
stdev <- sd(MLC_stratified_biomass$BIOMASS)
stderr <- sqrt(stdev^2/50)
#95% CI:
mean_biomass - 1.96 * stderr
mean_biomass + 1.96 * stderr
220.628 - 1.96 * stderr
220.628 + 1.96 * stderr

#95% CI:
mean_biomass - 2.58 * stderr
mean_biomass + 2.58 * stderr
220.628 - 2.58 * stderr
220.628 + 2.58  * stderr
MeanCI(MLC_stratified_biomass$BIOMASS, conf.level = 0.99)

#2'02"mins

library(dplyr)
group_by(MLC_stratified_biomass, RASTERVALU) %>%
  summarise(mean = mean(BIOMASS, na.rm = TRUE),
    sd = sd(BIOMASS, na.rm = TRUE) )

####################
####################
####################

Mean(Planted$BIOMASS)
MeanCI(Mixed_Broadleaved$BIOMASS)
MeanCI(Larch$BIOMASS)
MeanCI(Evergreen$BIOMASS)
MeanCI(MLC_stratified_biomass$BIOMASS)

summary(Planted$BIOMASS)
describe(Mixed_Broadleaved$BIOMASS)
describe(Larch$BIOMASS)
describe(Evergreen$BIOMASS)
describe(MLC_stratified_biomass$BIOMASS)

library(multcomp)
# Compute the analysis of variance among and between factored subsets
# One-way anova comparisons of means
res.aov <- aov(BIOMASS ~ RASTERVALU_factor, data = MLC_stratified_biomass)
summary(res.aov)
#Tukey multiple comparisons of means
#95% family-wise confidence level
summary(glht(res.aov, linfct = mcp(RASTERVALU_factor = "Tukey")))
kruskal.test(BIOMASS ~ RASTERVALU_factor, data = MLC_stratified_biomass)

#Canover rankings & anova test
```

```
ConoverTest(BIOMASS ~ RASTERVALU_factor, data = MLC_stratified_biomass)
describe(classest.forest$AREA)


#####################
#####################
#####################

library(ggpubr)
ggboxplot(my_data, x = "group", y = "weight",
          color = "group", palette = c("#00AFBB", "#E7B800", "#FC4E07"),
          order = c("ctrl", "trt1", "trt2"),
          ylab = "Weight", xlab = "Treatment")
ggline(my_data, x = "group", y = "weight",
       add = c("mean_se", "jitter"),
       order = c("ctrl", "trt1", "trt2"),
       ylab = "Weight", xlab = "Treatment")
# Box plot
boxplot(weight ~ group, data = my_data,
        xlab = "Treatment", ylab = "Weight",
        frame = FALSE, col = c("#00AFBB", "#E7B800", "#FC4E07"))
# plotmeans
library("gplots")
plotmeans(weight ~ group, data = my_data, frame = FALSE,
          xlab = "Treatment", ylab = "Weight",
          main="Mean Plot with 95% CI")
```

The following report addresses four questions from the exercise protocol and discusses the methods used. The aims of this exercise were to develop regression models for predicting AGB of the Clocaenog forest site and select the appropriate model candidate based on their internal error rate. Section 1 provides a summary of data and methods used to develop eight candidate regression models. The following section evaluates the candidate models using accuracy assessments. Important to note that integrating remote sensing data into a multi-phase forest inventory sampling requires a statistical framework that is based on unbiased model-assisted estimators (20–22). Therefore, efforts are made to check all four type (I-IV) errors and to evaluate if models are over-fitted to the data based on k-fold validation techniques.

1. Building a LiDAR model for predicting above ground biomass (AGB)

Models were calibrated between field plot data and remote sensing statistics, which were linked using global navigation satellite systems (23). Field data was collected from among 50 randomly sampled concentric circular plots (10m radius), including measurements of tree diameters at breast height (dbh, cm). Remote sensing LiDAR data was generated from an airborne laser scanning survey (ALS), from which three predictor variables were generated: mean tree height (MeanH) (11,24–29), standard deviation (SD), and fractional canopy height (Cover) based on ratio of height returns above 1.3m to total returns (30–35). Descriptive statistics of these predictor variables are presented below in Table 1.

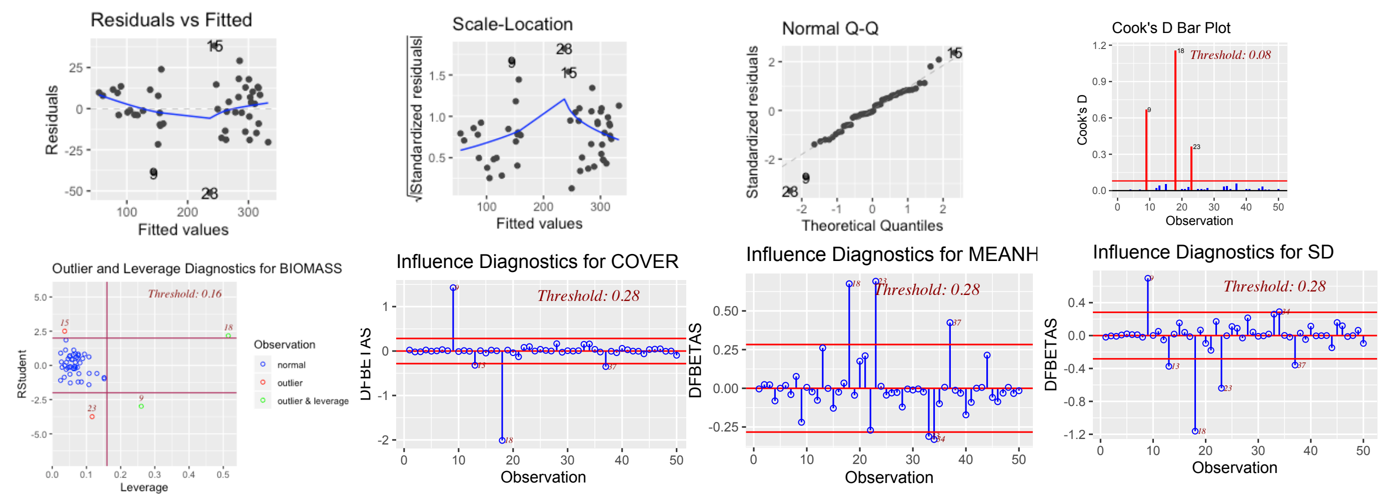
Table 1. Descriptives of predictors: Standard Deviation (SD), Standard Error (SE), Shapiro-Wilk (SW) score of normality.

Variable	Mean	SD	SE	Skewness	Kurtosis	SW
MeanH	13.05	5.77	0.82	-0.32	-1.40	***
Cover	0.92	0.08	0.01	-1.76	4.03	***
SD	5.41	2.84	0.40	-0.15	-1.47	***

A simple linear regression was conducted (M1) and its residuals were explored using regression diagnostics. These results are presented below in Table 2 and Figures 1 to 8. Results indicated signs of heteroscedasticity, multi-collinearity and influential outliers. Heteroscedasticity was observed by the Breusch-Pagan test ($p = 0.002$), and confirmed by the "Residuals vs Fitted" and "Scale Location". Although no collinearity problems were detected by Durba-Watson test, the Variance-Inflation-Factor identified Standard Deviation as a problematic predictor. The existence of influential outliers was observed using the Bonferroni test ($p = 0.025$). To explore measures of influence, analyses ran a Cook's Distance graph, a DFBETA panel, and a "Studentized Residuals vs Leverage" plot using the *lmtest* and *olsrr* packages in R. Two observations (obs 9 &18) were removed and a new dataset was developed "Clocaenog_Cleaned". To compare models, both datasets were used, and models corresponding to these are hereafter labeled "-clean" and "-clean".

Table 2. Hypothesis tests of linear regression: Shapiro-Wilk (SW) test of normality of residuals, Breusch-Pagan (BP) test of heteroskedasticity of residuals, Durbin-Watson (DW) test to search for multi-collinearity of data, Variance Inflation Factor (VIF) test to understand collinearity problems, Bonferroni (BF) test to search for outliers.

Model	SW	BP	DW	BF	Variance Inflation Factor		
					MeanH	SD	Cover
M1 (LM.comp)	0.131	0.002	0.345	0.025	3.234	5.451	2.886



Figures 1-8; Extensive diagnostics of M1 simple linear regression model

Based on these results, 7 new models were generated. M2 included a simple linear regression based on the cleaned data. M3 included a stepwise linear regression using two predictor variables (MeanH and SD) and complete dataset, which were selected from a stepwise backwards 10-kfold validation and the *caret* R package. M4 included a nonlinear least squares regression based on complete dataset and M5 included a nonlinear least squares regression using cleaned dataset. Using the *mass* R package, M6-8 models included iterated re-weighted least square regressions based on the Huber-loss estimator, the Tukey estimator and the Hampel estimator. Residuals of these 7 additional models were examined using same hypothesis tests as above. In Table 3 below, similar violations were observed. Due to its nonparametric nature, the model that showed fewer problems was M4.

Table 3. Regression diagnostics.

Model	SW	BP	DW	BF	Variance Inflation Factor		
					MeanH	SD	Cover
M2 (LM.clean)	0.949	0.187	0.303	0.001	3.743	4.951	9.493
M3 (LM.stepwise)	0.916	0.586	0.437	0.010	1.903	1.903	/
M4 (NLS.comp)	0.968	0.417	0.344	0.049	28.554	12.128	2.959
M5 (NLS.clean)	0.961	0.110	0.287	0.002	34.967	13.391	5.221
M6 (RR.huber)	0.002	0.002	0.345	0.025	3.234	5.451	2.886
M7 (RR.tukey)	0.000	0.002	0.345	0.025	3.234	5.451	2.886
M8 (RR.hampel)	0.005	0.002	0.345	0.025	3.234	5.451	2.886

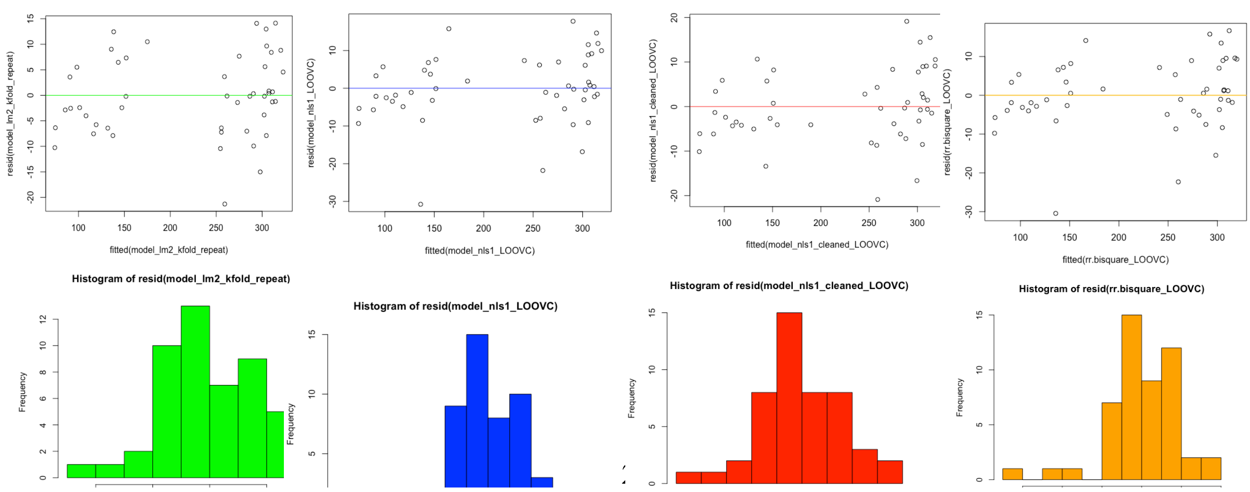
2. Accuracy assessments of LiDAR AGB models

As results above were not conclusive with regards to regression violations, all models were analysed using a resampling validation technique: the 10-Kfold cross validation (3-repeat). This method was used to provide estimates of the bias between resampling procedures and estimates of each model. It was also used to explore levels of precision in terms of the variation of results (RMSE). Using *DescTools* package in R, Theil's U estimate of error was provided to examine the level of unexplained variance in each model (Table 4.).

Table 4. Accuracy assessment of 8 candidate regressions using full models and cross validation estimates

Model	Full Models					Cross Validation	
	RMSE(%)	MAE(%)	Theil U_{error}	AIC	BIC	RMSE%	MAE%
1.LM.Full	16.44 (7.5)	11.95 (5.4)	0.998	427.727	437.286	16.63 (7.6)	13.67 (6.2)
2.LM.Cleaned	15.30 (6.9)	10.95 (4.9)	0.998	403.926	413.282	15.92 (7.1)	13.15 (5.9)
3.LM.Stepwise	17.86 (8.1)	12.47 (5.7)	0.997	435.041	442.689	16.44 (7.5)	13.34 (6.1)
4.NLS.Full	16.30 (7.4)	11.19 (5.1)	0.998	427.766	439.239	16.34 (7.4)	13.26 (6.0)
5.NLS.Cleaned	15.18 (6.8)	10.37 (4.6)	0.998	404.083	415.310	15.90 (7.1)	13.91 (6.2)
6.RR.Huber	13.11 (5.9)	11.17 (5.1)	1.001	430.087	429.647	17.19 (7.8)	14.08 (6.4)
7.RR.Tukey	12.14 (5.5)	11.17 (5.1)	1.006	435.415	444.975	17.31 (7.9)	13.94 (6.3)
8.RR.Hampel	13.34 (6.1)	11.17 (5.1)	1.001	428.938	438.499	17.48 (7.9)	14.08 (6.4)

Based on these results, the Tukey robust regression models provided highest accuracy based on its lowest RMSE score. However, all robust regressions showed signs of over-fitting the data. This was evident from the substantial difference observed between full model and cross validation error rates, as was as their higher AIC and BIC scores. Therefore, I selected the next best model M5: the nonlinear least squared model based on cleaned dataset (RMSE = 15.18). However, cleaning the dataset may not be sensible. As this remains uncertain, the residual spread of all four final trained candidate models are presented below (M2 = green, M4 = blue, M5 = red, M7 = orange).



Bibliography

1. Garcia M, Saatchi S, Ustin S, Balzter H. Modelling forest canopy height by integrating airborne LiDAR samples with satellite Radar and multispectral imagery. Int J Appl earth Obs Geoinf. 2018;66:159–73.

2. Guo Z, Chi H, Sun G. Estimating forest aboveground biomass using HJ-1 Satellite CCD and ICESat GLAS waveform data. Sci China Earth Sci. 2010;53(1):16–25.

3. Næsset E, Gobakken T, Solberg S, Gregoire TG, Nelson R, Ståhl G, et al. Model-assisted regional forest biomass estimation using LiDAR and InSAR as auxiliary data: A case study from a boreal forest area. Remote Sens Environ. 2011;115(12):3599–614.

4. Nelson R, Margolis H, Montesano P, Sun G, Cook B, Corp L, et al. Lidar-based estimates of aboveground biomass in the continental US and Mexico using ground, airborne, and satellite observations. Remote Sens Environ. 2017;188:127–40.

5. Nelson R, Krabill W, MacLean G. Determining forest canopy characteristics using airborne laser data. Remote Sens Environ. 1984;15(3):201–12.

6. Maclean GA, Krabill WB. Gross-merchantable timber volume estimation using an airborne LIDAR system. Can J Remote Sens. 1986;12(1):7–18.

7. Ene LT, Gobakken T, Andersen H-E, Næsset E, Cook BD, Morton DC, et al. Large-area hybrid estimation of aboveground biomass in interior Alaska using airborne laser scanning data. Remote Sens Environ. 2018;204:741–55.

8. Laurin GV, Puletti N, Chen Q, Corona P, Papale D, Valentini R. Above ground biomass and tree species richness estimation with airborne lidar in tropical Ghana forests. Int J Appl earth Obs Geoinf. 2016;52:371–9.

9. Magnussen S, Nord-Larsen T, Riis-Nielsen T. Lidar supported estimators of wood volume and aboveground biomass from the Danish national forest inventory (2012--2016). Remote Sens Environ. 2018;211:146–53.

10. Shao G, Shao G, Gallion J, Saunders MR, Frankenberger JR, Fei S. Improving Lidar-based aboveground biomass estimation of temperate hardwood forests with varying site productivity. Remote Sens Environ. 2018;204:872–82.

11. Næsset E. Predicting forest stand characteristics with airborne scanning laser using a practical two-stage procedure and field data. Remote Sens Environ. 2002;80(1):88–99.

12. Nelson R, Oderwald R, Gregoire TG. Separating the ground and airborne laser sampling phases to estimate tropical forest basal area, volume, and biomass. Remote Sens Environ. 1997;60(3):311–26.

13. Nilsson M. Estimation of tree heights and stand volume using an airborne lidar system. Remote Sens Environ. 1996;56(1):1–7.

14. Gougeon FA, Leckie DG, others. Forest information extraction from high spatial resolution images using an individual tree crown approach. Pacific Forestry Centre; 2002.

15. Reutebuch SE, Andersen H-E, McGaughey RJ. Light detection and ranging (LIDAR): an emerging tool for multiple resource inventory. J For. 2005;103(6):286–92.

16. Lee H, Slatton KC, Roth BE, Cropper Jr WP. Adaptive clustering of airborne LiDAR data to segment individual tree crowns in managed pine forests. Int J Remote Sens. 2010;31(1):117–39.

17. Andersen H-E, McGaughey RJ, Reutebuch SE. Estimating forest canopy fuel parameters using LIDAR data. Remote Sens Environ. 2005;94(4):441–9.

18. Shi Y, Wang T, Skidmore AK, Heurich M. Important LiDAR metrics for discriminating forest tree species in Central Europe. ISPRS J Photogramm Remote Sens. 2018;137:163–74.

19. Brandtberg T. Classifying individual tree species under leaf-off and leaf-on conditions using airborne lidar. ISPRS J Photogramm Remote Sens. 2007;61(5):325–40.

20. Tojal L-T, Bastarrika A, Barrett B, Sanchez Espeso JM, Lopez-Guede JM, Graña M. Prediction of Aboveground Biomass from Low-Density LiDAR Data: Validation over P. radiata Data from a Region North of Spain. Forests. 2019;10(9):819.

21. Corona P. Integration of forest mapping and inventory to support forest management. iForest-Biogeosciences For. 2010;3(3):59.

22. Corona P. Consolidating new paradigms in large-scale monitoring and assessment of forest ecosystems. Environ Res. 2016;144:8–14.

23. Valbuena R, Mauro F, Rodr\'\iguez-Solano R, Manzanera JA. Partial least squares for discriminating variance components in global navigation satellite systems accuracy obtained under scots pine canopies. For Sci. 2012;58(2):139–53.

24. Hopkinson C, Chasmer L, Hall RJ. The uncertainty in conifer plantation growth prediction from multi-temporal lidar datasets. Remote Sens Environ. 2008;112(3):1168–80.

25. Magnussen S, Boudewyn P. Derivations of stand heights from airborne laser scanner data with canopy-based quantile estimators. Can J For Res. 1998;28(7):1016–31.

26. Means JE, Acker SA, Fitt BJ, Renslow M, Emerson L, Hendrix CJ, et al. Predicting forest

stand characteristics with airborne scanning lidar. Photogramm Eng Remote Sensing. 2000;66(11):1367–72.

27. Naesset E. Determination of mean tree height of forest stands using airborne laser scanner data. ISPRS J Photogramm Remote Sens. 1997;52(2):49–56.

28. Næsset E, Økland T. Estimating tree height and tree crown properties using airborne scanning laser in a boreal nature reserve. Remote Sens Environ. 2002;79(1):105–15.

29. Popescu SC. Estimating biomass of individual pine trees using airborne lidar. Biomass and Bioenergy. 2007;31(9):646–55.

30. Hopkinson C, Chasmer L, Lim K, Treitz P, Creed I. Towards a universal lidar canopy height indicator. Can J Remote Sens. 2006;32(2):139–52.

31. Hopkinson C, Chasmer L. Testing LiDAR models of fractional cover across multiple forest ecozones. Remote Sens Environ. 2009;113(1):275–88.

32. Morsdorf F, Koetz B, Meier E, Itten KI, Allgöwer B. The potential of discrete return, small footprint airborne laser scanning data for vegetation density estimation. Proc ISPRS WG III/3. 2005;3(4):3.

33. Hill RA, Broughton RK. Mapping the understorey of deciduous woodland from leaf-on and leaf-off airborne LiDAR data: A case study in lowland Britain. ISPRS J Photogramm Remote Sens. 2009;64(2):223–33.

34. Wasser L, Day R, Chasmer L, Taylor A. Influence of vegetation structure on lidar-derived canopy height and fractional cover in forested riparian buffers during leaf-off and leaf-on conditions. PLoS One. 2013;8(1).

35. Morsdorf F, Kötz B, Meier E, Itten KI, Allgöwer B. Estimation of LAI and fractional cover from small footprint airborne laser scanning data based on gap fraction. Remote Sens Environ. 2006;104(1):50–61.

36. Cai T, Gao R, Hou J, Chen S, Wang D, He D, et al. A gram-gauss-newton method learning overparameterized deep neural networks for regression problems. arXiv Prepr arXiv190511675. 2019;

#5 steps that follow:
#1.Explore predictors and run linear model using full dataset. Run diagnostics & check violations in residuals
#2.Clean dataset removing biggest outliers
#3.Run 7 alternative models using full & cleaned dataset. Run diagnostics & accuracy assessment
#4.Choose model based on pre/post-diagnostics and accuracy assessment
#5.Plot residuals using benchmark diagnostics

R Syntax:

```
#### MSc Inventory Monitoring and Assessment ####
######  Assignment 3; LIDAR AGB Predict #########
############ S.Murphy 19-04-2020 ################

#using k-fold methods to estimate accuracy of model based on following:
#https://machinelearningmastery.com/how-to-estimate-model-accuracy-in-r-using-the-caret-package/
#library(caret)
#library(klaR)

library(readxl)
ClocaenogField <- read_excel("ClocaenogField.xlsx")
View(ClocaenogField)

#Explore data and normality of predictor variables:
describe(ClocaenogField$MEANH)
describe(ClocaenogField$COVER)
describe(ClocaenogField$SD)
shapiro.test(ClocaenogField$MEANH)
shapiro.test(ClocaenogField$COVER)
shapiro.test(ClocaenogField$SD)

#MODEL 1 (LM1): Linear regression
model_lm1 <- lm(BIOMASS ~ MEANH + COVER + SD, data = ClocaenogField)
summary (model_lm1)

#Run model diagnostics and check linear assumptions
ols_test_normality(model_lm1)
ols_test_breusch_pagan(model_lm1)
dwtest(model_lm1)
ols_coll_diag(model_lm1)
outlierTest(model_lm1, data=Duncan)
ols_test_outlier(model_lm1)
ols_vif_tol(model_lm1)
autoplot(model_lm1)
ols_plot_diagnostics(model_lm1)
ols_plot_cooksd_bar(model_lm1)
ols_plot_dfbetas(model_lm1)
ols_plot_resid_lev(model_lm1)
ols_plot_comp_plus_resid(model_lm1, print_plot = TRUE)

#Bonferroni results significant and influential outliers found.
#Create new dataframe removing outliers and re-run linear model.
#Check again for violations and compare model 1 & 2 with non-linear models.

#remove outliers from rows 9 and 18
```

```
ClocaenogField_cleaned <- ClocaenogField[-c(9,18),]

#MODEL 2 (LM_Cleaned): Linear regression using cleaned dataset
model_lm2 <- lm(BIOMASS ~ MEANH + COVER + SD, data = ClocaenogField_cleaned)
summary (model_lm2)

#run model diagnostics again
ols_test_normality(model_lm2)
ols_test_breusch_pagan(model_lm2)
dwtest(model_lm2)
ols_coll_diag(model_lm2)
outlierTest(model_lm2, data=Duncan)
ols_test_outlier(model_lm2)
ols_vif_tol(model_lm2)
autoplot(model_lm2)
ols_plot_diagnostics(model_lm2)
ols_plot_cooksd_bar(model_lm2)
ols_plot_dfbetas(model_lm2)
ols_plot_resid_lev(model_lm2)
ols_plot_comp_plus_resid(model_lm2, print_plot = TRUE)
#same violations found from bonferroni results and variance inflation factors
#resort to stepwise, non-linear and robust regression and compare...

#MODEL 3 (LM-STEP-Full): Stepwise Linear regression using full dataset
# Step-wise backwards 10-fold cross-validation test of model accuracy based on RMSE
# Set seed for reproducibility
# Set up repeated k-fold cross-validation
train.control_lm1 <- trainControl(method = "cv", number = 10)
# Train the model
step.model_lm1 <- train(BIOMASS ~ MEANH + COVER + SD, data = ClocaenogField, method =
"leapBackward", tuneGrid = data.frame(nvmax = 1:3), trControl = train.control_lm1)
step.model_lm1$results
summary(step.model_lm1$finalModel)
# Results suggest best 2-variable model contain MEANH and SD
# Step-wise model using most best 2-variables: MEANH and SD
model_lm_step_full <- lm(BIOMASS ~ MEANH + SD, data = ClocaenogField)
summary(model_lm_step_full)

#run diagnostics
ols_test_normality(model_lm_step_full)
ols_test_breusch_pagan(model_lm_step_full)
dwtest(model_lm_step_full)
ols_coll_diag(model_lm_step_full)
outlierTest(model_lm_step_full, data=Duncan)
ols_test_outlier(model_lm_step_full)
ols_vif_tol(model_lm_step_full)
autoplot(model_lm_step_full)
ols_plot_diagnostics(model_lm_step_full)
ols_plot_cooksd_bar(model_lm_step_full)
ols_plot_dfbetas(model_lm_step_full)
ols_plot_resid_lev(model_lm_step_full)
ols_plot_comp_plus_resid(model_lm_step_full, print_plot = TRUE)

# MODEL 4 (NLS) Non-linear least squares regression model using full dataset
# Non-linear least squares regression
```

```
model_nls1 <- lm(ClocaenogField$BIOMASS ~ ClocaenogField$MEANH + ClocaenogField$COVER + ClocaenogField$SD + I((ClocaenogField$MEANH + ClocaenogField$COVER + ClocaenogField$SD)^2))
summary(model_nls1)

# MODEL 5 (NLS) Non-linear least squares regression model using cleaned dataset
# Non-linear least squares regression
model_nls1_cleaned <- lm(ClocaenogField_cleaned$BIOMASS ~ ClocaenogField_cleaned$MEANH + ClocaenogField_cleaned$COVER + ClocaenogField_cleaned$SD + I((ClocaenogField_cleaned$MEANH + ClocaenogField_cleaned$COVER + ClocaenogField_cleaned$SD)^2))
summary(model_nls1_cleaned)

# MODEL 6 (RLM.Huber) Robust regression using the Huber M-estimator to reduce outlier influence. Though this does not reduce outliers in predictor variables, so Tukey needed below
rr.huber <- rlm(BIOMASS ~ MEANH + COVER + SD, data = ClocaenogField)
summary(rr.huber)
f.robftest(rr.huber,var = "MEANH")
f.robftest(rr.huber, var = "SD")
f.robftest(rr.huber, var = "COVER")


# MODEL 7 (RLM.Tukey) Robust regression using the Tukey M-estimator that assigns a weight of zero to influential outliers
rr.bisquare <- rlm(BIOMASS ~ MEANH + COVER + SD, data = ClocaenogField, psi = psi.bisquare)
summary(rr.bisquare)
f.robftest(rr.bisquare,var = "MEANH")
f.robftest(rr.bisquare, var = "SD")
f.robftest(rr.bisquare, var = "COVER")

# MODEL 8 (RLM.Tukey) Robust regression using the Tukey M-estimator that assigns a weight of zero to influential outliers
rr.hampel <- rlm(BIOMASS ~ MEANH + COVER + SD, data = ClocaenogField, psi = psi.hampel)
summary(rr.hampel)
f.robftest(rr.hampel,var = "MEANH")
f.robftest(rr.hampel, var = "SD")
f.robftest(rr.hampel, var = "COVER")


# Explore altnerative accuracy assessments:
# sidak p value adjustment
ols_test_breusch_pagan(model_nls1, rhs = TRUE, multiple = TRUE, p.adj = 'sidak')
# holm's p value adjustment
ols_test_breusch_pagan(model_nls1, rhs = TRUE, multiple = TRUE, p.adj = 'holm')
# Global test of model assumptions
library(gvlma)
gvmodel <- gvlma(model_lm1)
summary(gvmodel)
gvmodel_del <- deletion.gvlma(gvmodel)
summary(gvmodel_del)
plot(gvmodel_del)
display.delstats
summary.gvlmaDel
summary(gvmodel_del, allstats = FALSE)


gvmodel_lm2 <- gvlma(model_lm2)
```

```
summary(gvmodel_lm2)
gvmodel_del_lm2 <- deletion.gvlma(gvmodel_lm2)
summary(gvmodel_del_lm2)
plot(gvmodel_del_lm2)
display.delstats
summary.gvlmaDel
summary(gvmodel_del_lm2, allstats = FALSE)

gvmodel_lm1 <- gvlma(model_lm1_kfold)
summary(model_lm1_kfold)
gvmodel_del_lm2 <- deletion.gvlma(model_lm1_kfold)
summary(gvmodel_del_lm2)
plot(gvmodel_del_lm2)
display.delstats
summary.gvlmaDel
summary(gvmodel_del_lm2, allstats = FALSE)

#compare models
ols_mallows_cp(model_lm1, model_lm2)
ols_fpe(model_lm1)
ols_hsp(model_lm1)

ols_mallows_cp(model_lm1, model_lm2)
ols_mallows_cp(model_lm1, model_lm_step_full)
ols_mallows_cp(model_lm1, model_nls1)
ols_mallows_cp(model_lm1, model_nls1_cleaned)
ols_mallows_cp(model_lm1, rr.huber)
ols_mallows_cp(model_lm1, rr.bisquare)
ols_mallows_cp(model_lm1, rr.hampel)

AIC(model_lm1)
AIC(model_lm2)
AIC(model_lm_step_full)
AIC(model_nls1)
AIC(model_nls1_cleaned)
AIC(rr.huber)
AIC(rr.bisquare)
AIC(rr.hampel)

BIC(model_lm1)
BIC(model_lm2)
BIC(model_lm_step_full)
BIC(model_nls1)
BIC(model_nls1_cleaned)
BIC(rr.huber)
BIC(rr.bisquare)
BIC(rr.hampel)

glance(model_lm1)
glance(model_lm2)
glance(model_lm_step_full)
glance(model_nls1)
glance(model_nls1_cleaned)
glance(rr.huber)
glance(rr.bisquare)
```

```
glance(rr.hampel)

#Explore residual diagnostics
ols_plot_resid_box
ols_plot_resid_fit
ols_plot_resid_hist
ols_plot_resid_qq


#big outliers so choose to run a robust regression using MASS pkg:
#from https://stats.idre.ucla.edu/r/dae/robust-regression/
#Robust regression is iterated re-weighted least squares (IRLS). The
#command is rlm in the MASS package. There are several
#weighting functions that can be used for IRLS. We
#first use the Huber weights and then bi-square weighting. We
#will then look at the final weights created by the IRLS process.

#robust methods due to variance of resids and measures of influence:
#from https://stats.idre.ucla.edu/r/dae/robust-regression/
#We can see that the weight given to Mississippi
#is dramatically lower using the bisquare weighting
#function than the Huber weighting function and
#the parameter estimates from these two different
#weighting methods differ. When comparing the
#results of a regular OLS regression and a robust
#regression, if the results are very different,
#you will most likely want to use the results
#from the robust regression. Large differences
#suggest that the model parameters are being
#highly influenced by outliers. Different functions
#have advantages and drawbacks. Huber weights can
#have difficulties with severe outliers, and
#bisquare weights can have difficulties converging
#or may yield multiple solutions.

#(robust) sandwich variance estimator for linear regression:
#from: https://thestatsgeek.com/2014/02/14/the-robust-sandwich-variance-estimator-for-linear-regression-using-r/
#This method allowed us to estimate valid standard errors
#for our coefficients in linear regression, without requiring
#the usual assumption that the residual errors have constant variance.


# 10 k-fold cross validation
# define training control
train_control_kfold <- trainControl(method="cv", number=10)
metric <- "Accuracy"
# fix the parameters of the algorithm
grid <- expand.grid(.fL=c(0), .usekernel=c(FALSE))
# train the model
model_lm1_kfold <- train(BIOMASS ~ MEANH + SD + COVER, data=ClocaenogField, trControl=train_control_kfold)
# summarize results
print(model_lm1_kfold)
View(model_lm1_kfold)
```

```
summary(model_lm1_kfold)

set.seed(7)
accuracy_model_lm1 <- train(BIOMASS ~ MEANH + SD + COVER, data=ClocaenogField, metric=metric,
trControl=train_control_kfold)


# repeated 10 k-fold cross validation
# define training control
train_control_kfold_repeat <- trainControl(method="repeatedcv", number=10, repeats=3)
grid <- expand.grid(.fL=c(0), .usekernel=c(FALSE))
# train the model
model_lm1_kfold_repeat <- train(BIOMASS ~ MEANH + SD + COVER, data = ClocaenogField, trControl =
train_control_kfold_repeat)
model_lm2_kfold_repeat <- train(BIOMASS ~ MEANH + SD + COVER, data = ClocaenogField_cleaned,
trControl=train_control_kfold_repeat)
model_lm_step_full_kfold_repeat <- train(BIOMASS ~ MEANH + SD, data = ClocaenogField,
trControl=train_control_kfold_repeat)
model_nls1_kfold_repeat <- train(BIOMASS ~ MEANH + SD + COVER, data = ClocaenogField,
trControl=train_control_kfold_repeat)
model_nls1_cleaned_kfold_repeat <- train(BIOMASS ~ MEANH + SD + COVER, data =
ClocaenogField_cleaned, trControl=train_control_kfold_repeat)
rr.huber_kfold_repeat <- train(BIOMASS ~ MEANH + SD + COVER, data = ClocaenogField,
trControl=train_control_kfold_repeat)
rr.bisquare_kfold_repeat <- train(BIOMASS ~ MEANH + SD + COVER, data = ClocaenogField,
trControl=train_control_kfold_repeat)
rr.hampel_kfold_repeat <-train(BIOMASS ~ MEANH + SD + COVER, data = ClocaenogField,
trControl=train_control_kfold_repeat)

print(model_lm1_kfold_repeat)
print(model_lm2_kfold_repeat)
print(model_lm_step_full_kfold_repeat)
print(model_nls1_kfold_repeat)
print(model_nls1_cleaned_kfold_repeat)
print(rr.huber_kfold_repeat)
print(rr.bisquare_kfold_repeat)
print(rr.hampel_kfold_repeat)

# Leave-one-out cross validation
# define training control
train_control_LOOVC <- trainControl(method="LOOCV")
# train the model
model_lm1_LOOVC <- train(BIOMASS ~ MEANH + SD + COVER, data = ClocaenogField,
trControl=train_control_LOOVC)
model_lm2_LOOVC <- train(BIOMASS ~ MEANH + SD + COVER, data = ClocaenogField_cleaned,
trControl=train_control_LOOVC)
model_lm_step_full_LOOVC <- train(BIOMASS ~ MEANH + SD, data = ClocaenogField,
trControl=train_control_LOOVC)
model_nls1_LOOVC <- train(BIOMASS ~ MEANH + SD + COVER, data = ClocaenogField,
trControl=train_control_LOOVC)
model_nls1_cleaned_LOOVC <- train(BIOMASS ~ MEANH + SD + COVER, data =
ClocaenogField_cleaned, trControl=train_control_LOOVC)
rr.huber_LOOVC <- train(BIOMASS ~ MEANH + SD + COVER, data = ClocaenogField,
trControl=train_control_LOOVC)
```

```
rr.bisquare_LOOVC <- train(BIOMASS ~ MEANH + SD + COVER, data = ClocaenogField, trControl=train_control_LOOVC)
rr.hampel_LOOVC <-train(BIOMASS ~ MEANH + SD + COVER, data = ClocaenogField, trControl=train_control_LOOVC)

print(model_lm1_LOOVC)
summary(model_lm1_LOOVC)
View(model_lm1_LOOVC)

print(model_lm2_LOOVC)
print(model_lm_step_full_LOOVC)
print(model_nls1_LOOVC)
print(model_nls1_cleaned_LOOVC)
print(rr.huber_LOOVC)
print(rr.bisquare_LOOVC)
print(rr.hampel_LOOVC)

#run diagnostics
shapiro.test(resid(model_lm1_kfold_repeat))
shapiro.test(resid(model_lm2_kfold_repeat))
shapiro.test(resid(model_lm_step_full_kfold_repeat))
shapiro.test(resid(model_nls1_kfold_repeat))
shapiro.test(resid(model_nls1_cleaned_kfold_repeat))
shapiro.test(resid(rr.huber_kfold_repeat))
shapiro.test(resid(rr.bisquare_kfold_repeat))
shapiro.test(resid(rr.hampel_kfold_repeat))

bptest(model_lm1_kfold_repeat)
bptest(model_lm2_kfold_repeat)
bptest(model_lm_step_full_kfold_repeat)
bptest(model_nls1_kfold_repeat)
bptest(model_nls1_cleaned_kfold_repeat)
bptest(rr.huber_kfold_repeat)
bptest(rr.bisquare_kfold_repeat)
bptest(rr.hampel_kfold_repeat)

dwtest(rr.huber)
dwtest(rr.bisquare)
dwtest(rr.hampel)

outlierTest(lm(rr.huber))
outlierTest(lm(rr.bisquare))
outlierTest(lm(rr.hampel))

vif(rr.huber)
vif(rr.bisquare)
vif(rr.hampel)

AIC(model_lm1)
AIC(model_lm2)
AIC(model_lm_step_full)
AIC(model_nls1)
AIC(model_nls1_cleaned)
AIC(rr.huber)
AIC(rr.bisquare)
```

```
AIC(rr.hampel)


BIC(model_lm1)
BIC(model_lm2)
BIC(model_lm_step_full)
BIC(model_nls1)
BIC(model_nls1_cleaned)
BIC(rr.huber)
BIC(rr.bisquare)
BIC(rr.hampel)

TheilU(ClocaenogField$BIOMASS, model_lm1, type=1)
theil.wtd(model_lm1, weights = NULL)


bias_training <- rnorm(40, 2, sd = 0.5)
bias(bias_training, ClocaenogField)
bias(samp, pop)
bias(samp, pop, type = 'relative')
bias(samp, pop, type = 'standardized')
dev <- samp - pop
bias(dev)
nom.uncertainty(model_lm1)
lambda3(model_nls1)
CronbachAlpha(model_lm1)
guttman(model_lm1, missing = "complete", standardize = FALSE)


loq(model_lm1)



ols_plot_response(model_lm1)

u_theil_train <- createDataPartition(ClocaenogField$BIOMASS, p=0.5, list = FALSE)
u_theil_trainingData <- ClocaenogField[u_theil_train,]

u_theil_knn_model = train(model_lm1, data = u_theil_train, method = "knn", trControl = trainControl(method
= "cv", number = 5), tuneGrid = expand.grid((k = seq(1, 21, by = 2)))
u_theil_knn_model$modelType
summary(u_theil_knn_model)

u_theil_testData <- ClocaenogField[-u_theil_train,]
TheilU(actual_biomass, lm1_residuals)


dim(u_theil_trainingData)

lm1_residuals <-data.frame(lm=model_lm1$residuals)
training_sample <- rnorm (100, 2, sd = 0.5)
dev <- training_sample - ClocaenogField
bias(dev)
bias(mean(training_sample), ClocaenogField)
percent_bias(model_lm1, ClocaenogField)
```

```
lod(model_lm1, data = ClocaenogField)

summary(model_lm1)
computeBoundary(b1=2.8, b0=3.3, p=c(.5, .75))

m.nn_lm1      <- matchit(BIOMASS ~ MEANH + SD + COVER, data = ClocaenogField,      method=
"nearest",      ratio    = 1)
summary(m.nn)


ape(actual_biomass, lm1_residuals)


MAE(model_lm1)
MAE(model_lm2)
MAE(model_lm_step_full)
MAE(model_nls1)
MAE(model_nls1_cleaned)
MAE(rr.bisquare)
MAE(rr.bisquare)
MAE(rr.bisquare)

coxtest(model_lm1, model_lm1_kfold_repeat, data = ClocaenogField)

rsq.kl(model_lm1_kfold_repeat)
rsq.n(model_lm1_kfold_repeat)
rsq.partial(model_lm1_kfold_repeat)
rsq.sse(model_lm1_kfold_repeat)
rsq.v(model_lm1_kfold_repeat)

AIC(model_lm2)
AIC(model_lm_step_full)
AIC(model_nls1)
AIC(model_nls1_cleaned)
AIC(rr.huber)
AIC(rr.bisquare)
AIC(rr.hampel)


actual_biomass <- data.frame(ClocaenogField$BIOMASS)
actual_biomass_cleaned <- data.frame(ClocaenogField_cleaned$BIOMASS)


kfoldrepeat_residuals_lm1 <-data.frame(lm=model_lm1_kfold_repeat$residuals)
kfoldrepeat_residuals_lm2 <-data.frame(lm=model_lm2_kfold_repeat$residuals)
kfoldrepeat_residuals_lmstep <-data.frame(lm=model_lm_step_full_kfold_repeat$residuals)
kfoldrepeat_residuals_nls <-data.frame(lm=model_nls1_LOOVC$residuals)
kfoldrepeat_residuals_nlsclean <-data.frame(lm=model_nls1_cleaned_LOOVC$residuals)
kfoldrepeat_residuals_rrhuber <-data.frame(lm=rr.huber_LOOVC$residuals)
kfoldrepeat_residuals_rrbisq <-data.frame(lm=rr.bisquare_LOOVC$residuals)
kfoldrepeat_residuals_rrhampel <-data.frame(lm=rr.hampel_LOOVC$residuals)

lmFull_residuals <-data.frame(lm=model_lm1$residuals)
lmClean_residuals <-data.frame(lm=model_lm2$residuals)
```

```
lmStep_residuals <-data.frame(lm=model_lm_step_full$residuals)
nlsFull_residuals <-data.frame(lm=model_nls1$residuals)
nlsClean_residuals <-data.frame(lm=model_nls1_cleaned$residuals)
rrHuber_residuals <-data.frame(lm=rr.huber$residuals)
rrTukey_residuals <-data.frame(lm=rr.bisquare$residuals)
rrHampel_residuals <-data.frame(lm=rr.hampel$residuals)


library(DescTools)
TheilU(actual_biomass, lmFull_residuals)
TheilU(actual_biomass_cleaned, lmClean_residuals)
TheilU(actual_biomass, lmStep_residuals)
TheilU(actual_biomass, nlsFull_residuals)
TheilU(actual_biomass_cleaned, nlsClean_residuals)
TheilU(actual_biomass, rrHuber_residuals)
TheilU(actual_biomass, rrTukey_residuals)
TheilU(actual_biomass, rrHampel_residuals)

TheilU(actual_biomass, kfoldrepeat_residuals_lm1)
TheilU(actual_biomass_cleaned, kfoldrepeat_residuals_lm2)
TheilU(actual_biomass, kfoldrepeat_residuals_lmstep)
TheilU(actual_biomass, kfoldrepeat_residuals_nls)
TheilU(actual_biomass_cleaned, kfoldrepeat_residuals_nlsclean)
TheilU(actual_biomass, kfoldrepeat_residuals_rrhuber)
TheilU(actual_biomass, kfoldrepeat_residuals_rrbisq)
TheilU(actual_biomass, kfoldrepeat_residuals_rrhampel)

shapiro.test(resid(model_lm1))
shapiro.test(resid(model_lm2_kfold_repeat))
shapiro.test(resid(model_lm_step_full_kfold_repeat))
shapiro.test(resid(model_nls1_LOOVC))
shapiro.test(resid(model_nls1_cleaned_LOOVC))
shapiro.test(resid(rr.huber_LOOVC))
shapiro.test(resid(rr.bisquare_LOOVC))
shapiro.test(resid(rr.hampel_LOOVC))

bptest(model_lm1)
bptest(model_lm2_kfold_repeat)
bptest(resid(model_lm_step_full_kfold_repeat))
bptest(resid(model_nls1_LOOVC))
bptest(resid(model_nls1_cleaned_LOOVC))
bptest(resid(rr.huber_LOOVC))
bptest(resid(rr.bisquare_LOOVC))
bptest(resid(rr.hampel_LOOVC))


#model diagnostics:
#check for trends
#check for equal distribution of predictors
plot (ClocaenogField$MEANH, resid (model_lm1))
abline (0,0)
plot (ClocaenogField$MEANH, resid (model_lm1))
abline (0,0)
```

```
plot (ClocaenogField$COVER, resid (model2))
abline (0,0)
plot (ClocaenogField$SD, resid (model2))
abline (0,0)



PlotCandlestick(model_lm1)

#homoskedasticity
#predicted values aginst fitted
PlotQQ(model_lm1)
autoplot(model_lm1_kfold_repeat, label.size = 3)

plot(fitted(model_lm1_kfold_repeat), resid(model_lm1_kfold_repeat))
abline(a = coef(model_lm1_kfold_repeat), b = 0, col="green")
plot(fitted(model_nls1_LOOVC), resid(model_nls1_LOOVC))
lines(abline(a = coef(model_nls1_LOOVC), b = 0, col="red"))
plot(fitted(model_nls1_cleaned_LOOVC), resid(model_nls1_cleaned_LOOVC))
lines(abline(a = coef(model_nls1_cleaned_LOOVC), b = 0, col="orange"))
plot(fitted(rr.bisquare_LOOVC), resid(rr.bisquare_LOOVC))
lines(abline(a = coef(rr.bisquare_LOOVC), b = 0, col="yellow"))


plot(fitted(model_lm2_kfold_repeat), resid(model_lm2_kfold_repeat))
lines(abline(a = coef(model_lm2_kfold_repeat), b = 0, col="blue"))

legend(46, 15, legend = c("model1: linear", "model2: poly x^2", "model3: poly x^2 + x^3"),
      col=c("green", "blue", "red", "orange", "yellow"), lwd=3, bty="n", cex=0.9)
lines(smooth.spline(SBI_species$dbh_cm, predict(predict3.z05.55)), col="green", lwd=3, lty=3)


legend(46, 15, legend = c("model1: linear", "model2: poly x^2", "model3: poly x^2 + x^3"),
      col=c("red", "blue", "green"), lwd=3, bty="n", cex=0.9)



lines(smooth.spline(SS_species$dbh_cm, predict(predict2_C14.16)), lwd=3, col="blue")
predict3_C14.16 <- lm(SS_species$tree_agb_kg_C14.16 ~ SS_species$dbh_cm + I(SS_species$dbh_cm^2) +
I(SS_species$dbh_cm^3))
lines(smooth.spline(SS_species$dbh_cm, predict(predict3_C14.16)), col="green", lwd=3, lty=3)
legend(46, 15, legend = c("model1: linear", "model2: poly x^2", "model3: poly x^2 + x^3"),
#breusch-pagan test of homoskedasticity
bptest(model2)
bptest(model1)



#normality of residuals
qqnorm  (rstandard(model_lm1))
abline(0,1)
hist(resid(model2))
ols_test_normality(model2)
```

```
plot(data$BIOMASS, fitted(modmel2)
    abline(0,1)

    sqrt(mean(sqres))
    sqrt(mean(sqres))/mean(data$BIOMASS)*100



plot(SS_species$dbh_cm, SS_species$tree_agb_kg_C14.16)
abline(predict1_C14.16, col="red", lwd=3)
predict2_C14.16 <- lm(SS_species$tree_agb_kg_C14.16 ~ SS_species$dbh_cm + I(SS_species$dbh_cm^2))
lines(smooth.spline(SS_species$dbh_cm, predict(predict2_C14.16)), lwd=3, col="blue")
predict3_C14.16 <- lm(SS_species$tree_agb_kg_C14.16 ~ SS_species$dbh_cm + I(SS_species$dbh_cm^2) +
I(SS_species$dbh_cm^3))
lines(smooth.spline(SS_species$dbh_cm, predict(predict3_C14.16)), col="green", lwd=3, lty=3)
 <-data.frame(lm=model_lm1_kfold_repeat$residuals)


kfoldrepeat_residuals_lm2 <-data.frame(lm=model_lm2_kfold_repeat$residuals)
kfoldrepeat_residuals_lmstep <-data.frame(lm=model_lm_step_full_kfold_repeat$residuals)
kfoldrepeat_residuals_nls <-data.frame(lm=model_nls1_LOOVC$residuals)
kfoldrepeat_residuals_nlsclean <-data.frame(lm=model_nls1_cleaned_LOOVC$residuals)
kfoldrepeat_residuals_rrhuber <-data.frame(lm=rr.huber_LOOVC$residuals)
kfoldrepeat_residuals_rrbisq <-data.frame(lm=rr.bisquare_LOOVC$residuals)
kfoldrepeat_residuals_rrhampel <-data.frame(lm=rr.hampel_LOOVC$residuals)

shapiro.test(resid(model_lm1))
bptest(model_lm2_kfold_repeat)
bptest(model_lm_step_full_kfold_repeat)
bptest(model_nls1_kfold_repeat)
bptest(model_nls1_cleaned_LOOVC)
bptest(rr.huber_LOOVC)
bptest(rr.bisquare_LOOVC)
bptest(rr.hampel_LOOVC)

dwtest(rr.huber)
dwtest(rr.bisquare)
dwtest(rr.hampel)

outlierTest(lm(rr.huber))
outlierTest(lm(rr.bisquare))
outlierTest(lm(rr.hampel))


plot(fitted(model_lm2_kfold_repeat), resid(model_lm2_kfold_repeat))
abline(0,0, col = "green")
hist(resid(model_lm2_kfold_repeat), col = "green")
```

```
plot(fitted(model_nls1_LOOVC), resid(model_nls1_LOOVC))
abline(0,0, col = "blue")
hist(resid(model_nls1_LOOVC), col = "blue")

plot(fitted(model_nls1_cleaned_LOOVC), resid(model_nls1_cleaned_LOOVC))
abline(0,0, col = "red")
hist(resid(model_nls1_cleaned_LOOVC), col = "red")

plot(fitted(rr.bisquare_LOOVC), resid(rr.bisquare_LOOVC))
abline(0,0, col = "orange")
hist(resid(rr.bisquare_LOOVC), col = "orange")
```